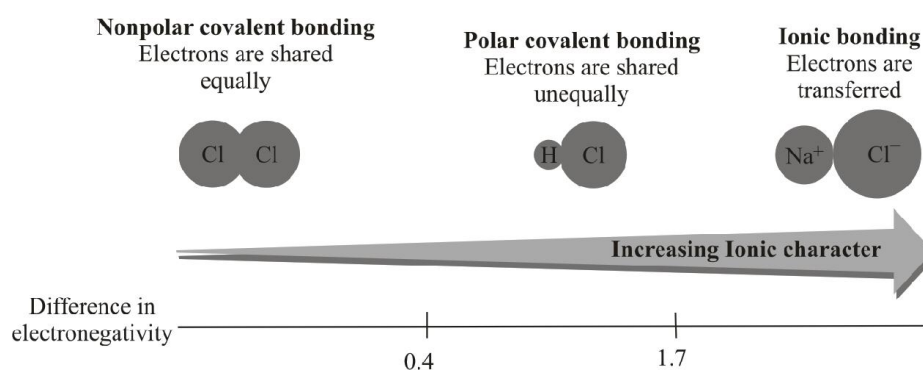


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CHEMICAL BONDING

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THEORY

1. INTRODUCTION

Atoms are usually not capable of free existence but groups of atoms of the same or different elements exist as one species, e.g., H_2 , O_2 , P_4 , S_8 , H_2O .

A group of atoms existing together as one species and having characteristic properties is called a molecule.

Obviously, there must be some force which holds these atoms together within the molecules.

2. CHEMICAL BOND

This force which holds the atoms together within a molecule is called a chemical bond.

2.1 Why do atoms combine ?

Lewis-Kossel Approach to Chemical Bonding

The atoms of different elements combine with each other in order to complete their respective octets (i.e., 8 electrons in their outermost shell) or duplet (i.e., outermost shell having 2 electrons) in case of H, Li and Be to attain stable nearest noble gas configuration.

2.2 Modes of Chemical Composition

This can occur in two ways :

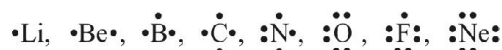
1. By complete transference of one or more electrons from one atom to another. This process is referred to as electrovalency and the chemical bond formed is termed as electrovalent bond or ionic bond.
2. By sharing of electrons. This can occur in two ways as follows :
 - (a) When the shared electrons are contributed by the two combining atoms equally, the bond formed is called covalent bond.
 - (b) When these electrons are contributed entirely by one of the atoms but shared by both, the bond formed is known as a coordinate bond, also called dative bond.

3. LEWIS SYMBOLS

In the formation of a molecule, only the outer shell electrons are involved and they are known as valence electrons. The inner shell electrons are well protected and are generally not involved in the combination process. It is, therefore, quite reasonable to consider the outer shell electrons, i.e., valence shell electrons while discussing chemical bonds.

G.N. Lewis introduced simple symbols to denote the valence shell electrons in an atom. The outer shell electrons are shown as dots surrounding the symbol of the atom. These symbols are known as Lewis symbols or electron dot

symbols. These symbols ignore the inner shell electrons. A few examples are given below :

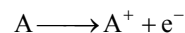


4. IONIC BOND

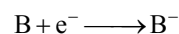
When a bond is formed by complete transference of electrons from one atom to another so as to complete their outermost orbits by acquiring 8 electrons (i.e., octet) or 2 electrons (i.e., duplet) in case of hydrogen, lithium etc. and hence acquire the stable nearest noble gas configuration, the bond formed is called ionic bond or electrovalent bond.

4.1 Explanation of Ionic Bond

Atoms are electrically neutral. Therefore, they possess equal number of protons and electrons. On losing an electron, an atom becomes positively charged since now the number of protons exceeds the number of electrons.

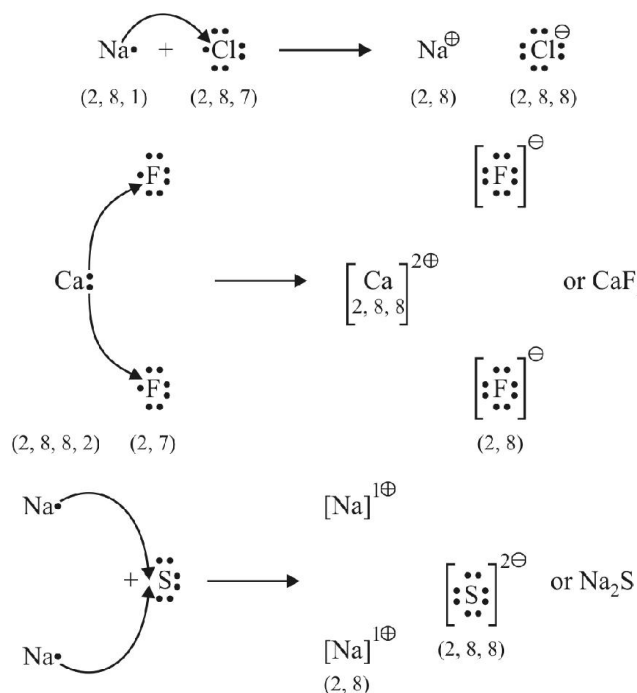


On the other hand, in case of atom, gaining the electron, the number of electrons exceeds the number of protons and thus the atom becomes negatively charged.



The oppositely charged particles formed above attract each other by electrostatic forces of attraction. The bond thus formed is known as electrovalent or ionic bond.

Examples



4.2 Electrovalency

The number of electrons lost or gained during the formation of an electrovalent linkage is termed as the electrovalency of the element.

For example, sodium and calcium lost 1 and 2 electrons respectively and so their valencies are 1 and 2. Similarly, chlorine and oxygen gain 1 and 2 electrons respectively, so they possess an electrovalency of 1 and 2. In other words, valency is equal to the charge on the ion.

4.3 Factors governing the formation of ionic bonds**(i) Ionisation Enthalpy (Ionization Energy)**

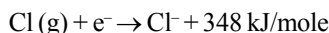
Ionisation enthalpy of any element is the amount of energy required to remove an electron from the outermost shell of an isolated atom in gaseous phase so as to convert it into a gaseous positive ion.

It is clear that lesser the ionisation enthalpy, easier will be the removal of an electron, i.e., formation of a positive ion and hence greater the chances of formation of an ionic bond. Ionisation enthalpy (I.E.) of alkali metals (i.e., group I elements) is low, hence they have more tendency to form positive ions.

(ii) Electron Gain Enthalpy (Electron Affinity)

Electron affinity or Electron gain enthalpy of an element is the enthalpy change that takes place when an extra electron is added to an isolated atom in the gaseous phase to form a gaseous negative ion.

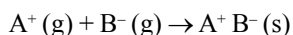
Higher is the electron affinity, more is the energy released and stabler will be the negative ion produced. Consequently, the probability of formation of ionic bond will be enhanced. Halogens possess high electron affinity. So the formation of their negative ions is very common, e.g., in case of chlorine, electron affinity is +348 kJ/mole, i.e.,



or E.Z. = + 348 kJ mol⁻¹

(iii) Lattice Enthalpy (Lattice Energy)

In the formation of ionic compounds, the positively charged ions combine with negatively charged ions to form the compound.



The energy released when the requisite number of gaseous positive and negative ions combine to form one mole of the ionic compound is called lattice enthalpy.

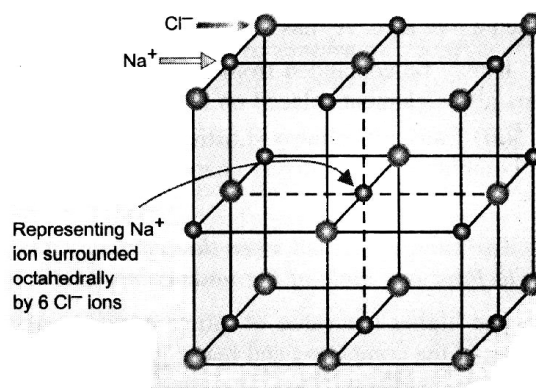
4.4 Characteristics of Ionic Compounds**1. Physical State**

These compounds usually exist in the solid state.

2. Crystal Structure

X-ray analysis of the ionic compounds shows that they exist as ions and not as molecules. These ions are arranged in a regular pattern in the three dimensional space to form a lattice.

The pattern of arrangement, however, depends upon the size and charges of the ions. For example, in case of sodium chloride, each sodium ion is surrounded by six chloride ions and each chloride by six sodium ions, thus giving rise to a three dimensional octahedral crystal structure (figure). The formula of an ionic compound merely indicates the relative number of ions present.



Crystal structure of NaCl

3. High melting and boiling points

Ionic compounds possess high melting and boiling points. This is because ions are tightly held together by strong electrostatic forces of attraction and hence a huge amount of energy is required to break the crystal lattice.

4. Solubility

Electrovalent compounds are soluble in solvents like water which are polar in nature and have high dielectric constant. It is due to the reason that the polar solvent interacts with the ions of the crystals and further the high dielectric constant of the solvent (i.e., capacity of the solvent to weaken the forces of attraction) cuts off the force of attraction between these ions. Furthermore, the ions may combine with the solvent to liberate energy called the hydration enthalpy which is sufficient to overcome the attractive forces between the ions.

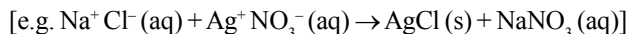
Non-polar solvents like carbon tetrachloride, benzene etc. having low dielectric constants are not capable of dissolving ionic solids. Hence, ionic solids are soluble in polar solvents and insoluble in non-polar solvents.

5. Electrical conductivity

Ionic compounds are good conductors of electricity in solution or in the molten state. In solution or molten state, their ions are free to move. As the ions are charged, they are attracted towards electrodes and thus act as carriers of electric current.

6. Ionic Reactions

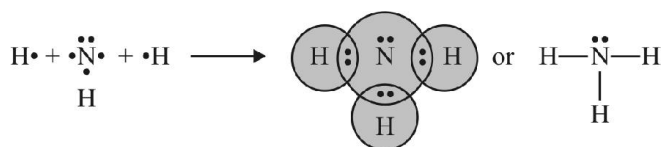
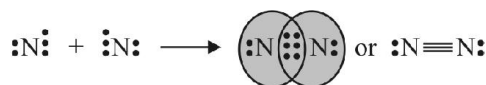
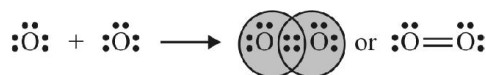
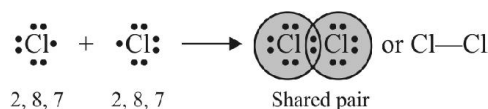
The reactions of the ionic compounds are, in fact, the reactions between the ions produced in solution. As the oppositely charged ions combine quickly, these reactions are, therefore, quite fast.



5. COVALENT BOND

The bond formed between the two atoms by mutual sharing of electrons between them so as to complete their octets or duplets in case of elements having only one shell is called covalent bond or covalent linkage and the number of electrons contributed by each atom is known as covalency.

Example



Example

Draw the Lewis dot structure of HCN molecule.

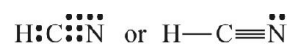
Sol. Step-1 : Total number of valence electrons in HCN = $1 + 4 + 5 = 10$ (${}_1\text{H} = 1, {}_6\text{C} = 2, 4, {}_7\text{N} = 2, 5$)

Step-2 : Skeletal structure is HCN (C is least electronegative).

Step-3 : Putting one shared pair of electrons between H and C and one between C and N, and the remaining as lone pairs, we have



In this structure, duplet of H is complete but octets of C and N are not complete. Hence, multiple bonding is required between C and N. Octets of C and N will be complete if there is triple bond between C and N. Thus,



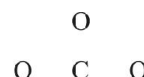
Example

Draw the Lewis dot structure of CO_3^{2-} ion.

Sol. Step-1 : Total number of valence electrons of CO_3^{2-}
 $= 4 + 3 \times 6 = 22$ (${}_6\text{C} = 2, 4, {}_8\text{O} = 2, 6$)

Step-2 : Total number of electrons to be distributed in CO_3^{2-}
 $= 22 + 2$ (for two units -ve charge) = 24

Step-3 : The skeletal structure of CO_3^{2-} is



Step-4 : Putting one shared pair of electrons between each C and O and completing the octets of oxygen, we have



In this structure, octet of C is not complete. Hence, multiple bonding is required between C and one of the O-atoms. Drawing a double bond between C and one O-atom serves the purpose :

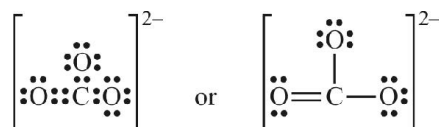


Table : Lewis structures of some typical molecules and ions

Molecule/ Ion		Lewis Representation		
(i)	CO	$\text{:C}\equiv\text{O:}$	or	$\text{:C}\equiv\text{O:}$
(ii)	CO ₂	$\text{:O}::\text{C}::\text{:O}$	or	$\text{:O}=\text{C}=\text{:O}$
(iii)	O ₂	$\text{:O}::\text{:O}$	or	$\text{:O}=\text{O:}$
(iv)	O ₃	$\begin{array}{c} \text{:O:}^+ \\ \diagup \quad \diagdown \\ \text{:O:} \quad \text{:O:}^- \end{array}$	or	$\begin{array}{c} \text{:O:}^+ \\ \diagup \quad \diagdown \\ \text{:O:} \quad \text{:O:}^- \end{array}$
(v)	H ₂ O ₂	$\text{H}:\text{:O}::\text{:O}::\text{H}$	or	$\text{H}-\text{:O}-\text{:O}-\text{H}$
(vi)	SO ₃	$\begin{array}{c} \text{:O:} \\ \diagup \quad \diagdown \\ \text{:O}::\text{S}::\text{:O:} \\ \text{:O:} \end{array}$	or	$\begin{array}{c} \text{:O:} \\ \diagup \quad \diagdown \\ \text{:O}=\text{S}-\text{:O:} \\ \text{:O:} \end{array}$
(vii)	HClO ₄ (Perchloric acid)	$\begin{array}{c} \text{:O:} \\ \diagup \quad \diagdown \\ \text{H}:\text{:O}::\text{Cl}::\text{:O:} \\ \text{:O:} \end{array}$	or	$\begin{array}{c} \text{:O:} \\ \diagup \quad \diagdown \\ \text{H}-\text{:O}-\text{Cl}-\text{:O:} \\ \text{:O:} \end{array}$
(viii)	HNO ₂ (Nitrous acid)	$\text{H}:\text{:O}::\text{N}::\text{:O:}$	or	$\text{H}-\text{:O}-\text{N}=\text{:O:}$
(ix)	HNO ₃ (Nitric acid)	$\begin{array}{c} \text{:O:} \\ \diagup \quad \diagdown \\ \text{H}:\text{:O}::\text{N} \\ \text{:O:} \end{array}$	or	$\begin{array}{c} \text{:O:} \\ \diagup \quad \diagdown \\ \text{H}-\text{:O}-\text{N} \\ \text{:O:} \end{array}$
(x)	H ₃ PO ₄ (Phosphoric acid)	$\begin{array}{c} \text{:O:} \text{H} \\ \diagup \quad \diagdown \\ \text{H}:\text{:O}::\text{P}::\text{:O:} \text{H} \\ \text{:O:} \end{array}$	or	$\begin{array}{c} \text{:O:} \text{H} \\ \diagup \quad \diagdown \\ \text{H}-\text{:O}-\text{P}-\text{:O:} \text{H} \\ \text{:O:} \end{array}$
(xi)	CN ⁻ (Cyanide ion)	$[\text{:C}::\text{N:}]^-$	or	$[\text{:C}\equiv\text{N:}]^-$
(xii)	NO ₂ ⁻ (Nitrite ion)	$[\text{:O}::\text{N}::\text{:O:}]^-$	or	$[\text{:O}=\text{N}-\text{:O:}]^-$
(xiii)	NO ₃ ⁻ (Nitrate ion)	$\begin{array}{c} \text{:O:} \\ \diagup \quad \diagdown \\ \text{:O}::\text{N}::\text{:O:} \\ \text{:O:} \end{array}$	or	$\begin{array}{c} \text{:O:} \\ \diagup \quad \diagdown \\ \text{:O}=\text{N}-\text{:O:} \\ \text{:O:} \end{array}$

Table : Lewis structures of some typical molecules and ions

Molecule/ Ion		Lewis Representation		
(xiv)	SO ₄ ²⁻ (Sulphate ion)	$\left[\begin{array}{c} \text{:O:} \\ \diagup \quad \diagdown \\ \text{:O}::\text{S}::\text{:O:} \\ \text{:O:} \end{array} \right]^{2-}$	or	$\left[\begin{array}{c} \text{:O:} \\ \diagup \quad \diagdown \\ \text{:O}-\text{S}-\text{:O:} \\ \text{:O:} \end{array} \right]^{2-}$
(xv)	O ₂ ²⁻ (Peroxide ion)	$[\text{:O}::\text{:O:}]^{2-}$	or	$[\text{:O}-\text{:O:}]^{2-}$
(xvi)	H ₃ O ⁺ (Hydronium ion)	$\left[\begin{array}{c} \text{H}:\text{:O}::\text{H} \\ \text{H} \end{array} \right]^+$	or	$\left[\begin{array}{c} \text{H}-\text{:O}-\text{H} \\ \text{H} \end{array} \right]^+$

6. FORMAL CHARGE

The formal charge on an atom in a molecule or ion is defined as the difference between the number of valence electrons of that atom in the free state and the number of electrons assigned to that atom in the Lewis structure, assuming that in each shared pair of electrons, the atom has one electron of its own and the lone pair on it belongs to it completely. Thus, it can be calculated as follows :

$$\left[\begin{array}{l} \text{Formal charge (F.C.) on} \\ \text{an atom in a molecule/ion} \end{array} \right] = \left[\begin{array}{l} \text{Total no. of valence electrons} \\ \text{in the free atom} \end{array} \right] -$$

$$\left[\begin{array}{l} \text{Total no. of electrons of} \\ \text{lone pairs (non - bonding electrons)} \end{array} \right] - \frac{1}{2}$$

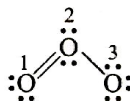
$$\left[\begin{array}{l} \text{Total no. of shared electrons} \\ \text{(bonding electrons)} \end{array} \right]$$

$$\text{i.e., FC} = V - L - \frac{1}{2}S$$

Example

Calculate formal charge on each O-atom of O_3 molecule.

Sol. Lewis structure of O_3 is :



The atoms have been numbered as 1, 2 and 3.

Formal charge on end O-atom numbered 1

$$= 6 - 4 - \frac{1}{2}(4) = 0$$

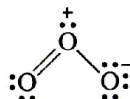
Formal charge on central O-atom numbered 2

$$= 6 - 2 - \frac{1}{2}(6) = +1$$

Formal charge on end O-atom numbered 3

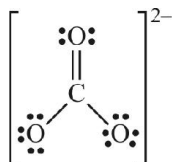
$$= 6 - 6 - \frac{1}{2}(2) = -1$$

Hence, we represent O_3 along with formal charges as :

**Example**

Write the formal charges on atoms in (i) carbonate ion (ii) nitrite ion.

Sol. (i) Lewis structure of CO_3^{2-} ion is



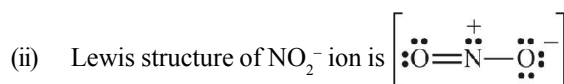
$$\text{Formal charge on C atom} = 4 - 0 - \frac{1}{2}(8) = 0,$$

Formal charge on double bonded O atom

$$= 6 - 4 - \frac{1}{2}(4) = 0$$

Formal charge on single bonded O atom

$$= 6 - 6 - \frac{1}{2}(2) = -1$$



Formal charge on N atom

$$= 5 - 2 - \frac{1}{2}(6) = 0,$$

Formal charge on double bonded O atom

$$= 6 - 4 - \frac{1}{2}(4) = 0$$

Formal charge on single bonded O atom

$$= 6 - 6 - \frac{1}{2}(2) = -1$$

Significance of formal charge. The main advantage of the calculation of formal charges is that it helps to select the most stable structure, i.e., the one with least energy out of the different possible Lewis structures. The most stable is the one which has the smallest formal charges on the atoms.

7. LIMITATIONS OF LEWIS-LANGMUIR CONCEPT OF COVALENT BOND

Lewis-Langmuir concept of covalent bond, as explained in ec. 4.8, has the following limitations :

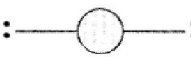
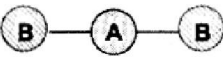
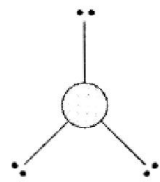
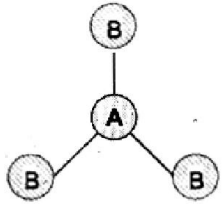
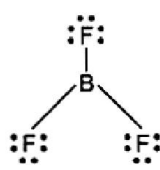
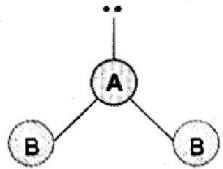
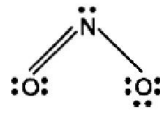
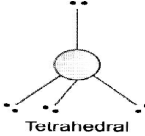
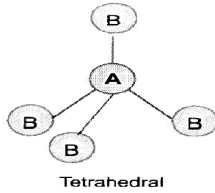
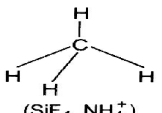
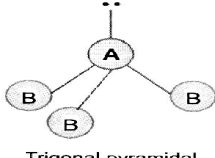
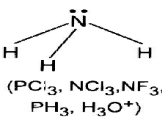
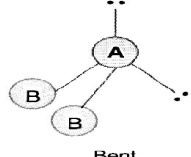
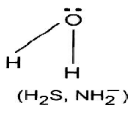
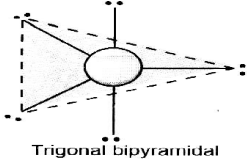
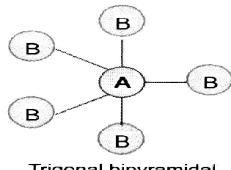
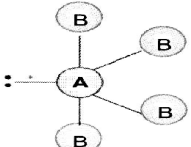
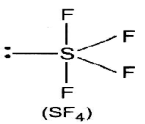
- It could not explain how the atoms are held together in the molecules like H_2 , Cl_2 etc. in which there are no ions and hence there are no electrostatic forces of attraction, i.e., it could not explain the formation of a covalent bond.
- It could not explain the shapes of molecules containing covalent bonds.
- It could not explain the release of energy during the formation of a covalent bond.

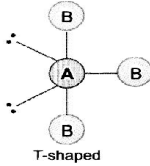
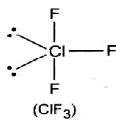
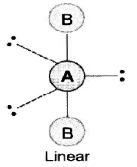
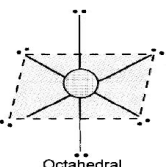
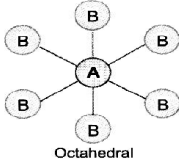
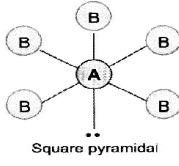
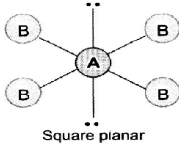
8. VALENC SHELL ELECTRON PAIR REPULSION THEORY

The first simple theory that was put forward to explain the shapes of molecules is known as Valence Shell. Electron Pair Repulsion (VSEPR) theory. This theory was given by Sidgwick and Powell in 1940 and was further improved by Nyholm and Gillespie in 1957.

The electron pairs surrounding the central atom repel one another and move so far apart from one another that there are no further repulsions between them. As a result, the molecule has minimum energy and maximum stability.

Table : Shapes (Geometry) of molecules containing bond pairs only or bond pairs and lone pairs

Total No. of electron pairs	Geometry of the electron pairs	Bond pairs	Lone pairs	Geometry (Shape) of the molecule	Illustrative Example (Other Examples)
		(Molecular Formula)			
2	 Linear	2	0	 Linear	$\ddot{\text{O}}=\text{C}=\ddot{\text{O}}$ (BeF ₂ , BeCl ₂ , HgCl ₂)
3	 Triangular planar	3	0	 Triangular planar	 (AlCl ₃ , SO ₃)
		(AB ₃)			
		2	1	 Bent (V-shape)	 (SO ₂ , O ₃ , SnCl ₂)
		(AB ₂ L)			
4	 Tetrahedral	4	0	 Tetrahedral	 (SiF ₄ , NH ₄ ⁺)
		3	1		
		(AB ₃ L)		 Trigonal pyramidal	 (PCl ₃ , NCl ₃ , NF ₃ , PH ₃ , H ₃ O ⁺)
		(AB ₂ L ₂)			
		2	2	 Bent	 (H ₂ S, NH ₂ ⁻)
		(AB ₂ L ₂)			
5	 Trigonal bipyramidal	5	0	 Trigonal bipyramidal	PCl ₅ (PF ₅ , SbCl ₅ , AsF ₅)
		(AB ₅)			
		4	1	 See saw	 (SF ₄)
		(AB ₄ L)			

		3	2		
		(AB ₃ L ₂)		T-shaped	
		2	3		XeF ₂ (I ₃ ⁻ , Br ₃ ⁻ , ICl ₂ ⁻)
		(AB ₂ L ₃)		Linear	
6		6	0		SF ₆
		(AB ₆)		Octahedral	
		5	1		ClF ₅ (IF ₅ , BrF ₅)
		(AB ₅ L)		Square pyramidal	
		4	2		XeF ₄
		(AB ₄ L ₂)		Square planar	

8.1 Calculation of total number of electron pairs, bond pairs and lone pairs and predicting the shapes of the molecules and ions.

- (i) Total no. of electron pairs around the central atom

$$= \frac{1}{2} (\text{No. of valence electrons of central atom} +$$

No. of atoms linked to central atom by single bonds)

For negative ions, add number of electrons equal to the units of negative charge on the ion to the valence electrons of the central atom.

For positive ions, subtract number of electrons equal to the units of positive charge on the ion from the valence electrons of the central atom.

- (ii) No. of bond pairs (shared pairs) = No. of atoms linked to central atom by single bonds.
(iii) No. of lone pairs = Total no. of electron pairs – No. of shared pairs.

Example

On the basis of VSEPR theory, predict the shapes of the following :

- (i) ClF₃ (ii) BrF₅

Sol. (i) Shape of ClF₃

No. of valence electrons of the central Cl atom = 7

No. of atoms linked to it by single bonds = 3.

$$\therefore \text{Total no. of electron pairs around Cl} = \frac{7+3}{2} = 5$$

No. of bond pairs = No. of atoms linked to Cl = 3.

$$\therefore \text{No. of lone pairs} = 5 - 3 = 2.$$

Thus, the molecule is of the type AB₃L₂.

Hence, it is T-shaped.

(ii) Shape of BrF₅

No. of valence electrons of central Br atom = 7

No. of atoms linked to it by single bonds = 5.

$$\therefore \text{Total no. of electron pairs around Br} = \frac{7+5}{2} = 6$$

No. of bond pairs = No. of atoms linked to Br = 5.

$$\therefore \text{No. of lone pairs} = 6 - 5 = 1.$$

Thus, the molecule is of the type AB₅L. Hence, it has square pyramidal shape.

9. HYBRIDISATION

Hybridisation is defined as the mixing of the atomic orbitals belonging to the same atom but having slightly different energies so that a redistribution of energy takes place between them resulting in the formation of new orbitals of equal energies and identical shapes. The new orbitals thus formed are known as hybrid orbitals.

Some Important Points about Hybridisation

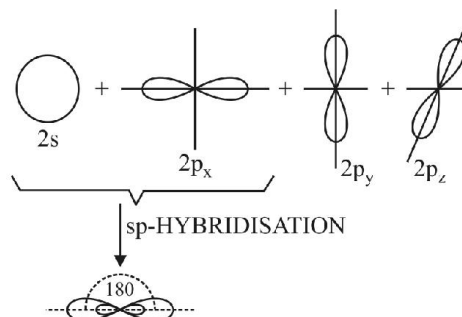
- Only those orbitals which have approximately equal energies and belong to the same atom or ion can undergo hybridisation.
- Number of hybrid orbitals produced is equal to the number of atomic orbitals mixed.
- It is not necessary that all the half-filled orbitals must participate in hybridisation. Similarly, it is not necessary that only half-filled orbitals should participate in hybridisation. Even completely filled orbitals with slightly different energies can also participate.
- Hybridisation never takes place in isolated atoms but it occurs only at the time of bond formation.
- Type of hybridisation indicates the geometry of molecules. One can tell the shape of a molecule by knowing the kind of hybridisation involved.
- The bigger lobe of the hybrid orbital always has +ve sign while the smaller lobe on the opposite side has a -ve sign.

9.1 Type of Hybridisation

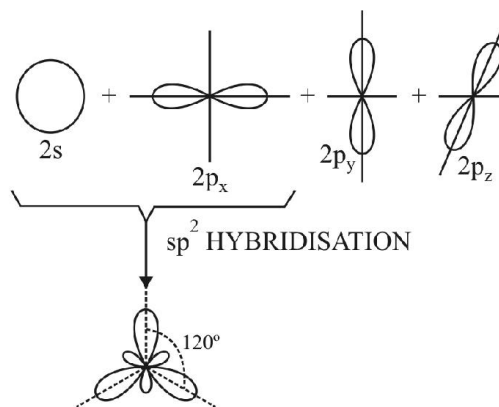
1. Diagonal or sp hybridisation

When one s and one p orbital belonging to the same main shell of an atom mix together to form two new equivalent orbitals, the type of hybridisation is called sp hybridisation

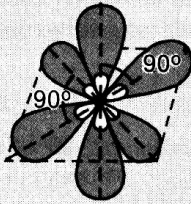
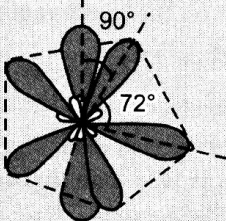
or diagonal hybridisation. The new orbitals formed are called sp hybrid orbitals.

TWO sp -HYBRID ORBITALS2. Trigonal or sp^2 hybridisation

When one s and two p orbitals of the same shell of an atom mix to form three new equivalent orbitals, the type of hybridisation is called sp^2 hybridisation or trigonal hybridisation. The new orbitals formed are called sp^2 hybrid orbitals.

THREE sp^2 HYBRID ORBITALS

Types of Hybridisation	Atomic orbitals involved	Representing directions of hybrid orbitals formed along with bond angles	Shape of the molecule	Examples
1. sp	one s + one p	180°	Linear	BeCl_2 , BeH_2 , C_2H_2 , HgCl_2
2. sp^2	one s + two p	120°	Triangular planar	BF_3 , BCl_3 , C_2H_4 , NO_3^- , CO_3^{2-}
3. sp^3	one s + three p	$109^\circ 28'$	Tetrahedral	CH_4 , CCl_4 , SnCl_4 , NH_4^+
4. dsp^2	one d ($d_{x^2-y^2}$) + one s + two p	90°	Square planar	$[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{PtCl}_4]^{2-}$
5. sp^3d	one s + three p + one d (d_{z^2})	90° , 120°	Trigonal bipyramidal	PF_5 , PCl_5

Types of Hybridisation	Atomic orbitals involved	Representing directions of hybrid orbitals formed along with bond angles	Shape of the molecule	Examples
6. sp^3d^2 or d^2sp^3	one s + three p + two d ($d_{x^2-y^2}$ and d_{z^2})		Octahedral	SF_6 , $[CrF_6]^{3-}$, $[Co(NH_3)_6]^{3+}$
7. sp^3d	one s + three p + d_{xy} , d_{yz} , d_{xz}		Pentagonal bipyramidal	IF_7

9.2 Predicting Hybridisation

(a) Predicting hybridisation

Calculate the number of hybrid orbitals (X) to be formed by the central atom as follows :

$$X = \frac{1}{2} \left[\left\{ \begin{array}{l} \text{No. of valence electrons} \\ \text{of the central atom} \end{array} \right\} + \right.$$

$$\left. + \left\{ \begin{array}{l} \text{Charge on the anion} \\ \text{if the given species is} \\ \text{a polyatomic anion} \end{array} \right\} \right]$$

$$\text{i.e., } X = \frac{1}{2} [VE + MA - c + a]$$

Note that only monovalent atoms (MA) or groups are to be considered. For divalent ions, $MA = 0$.

If $X = 2$, it means two hybrid orbitals are to be formed. Hence, hybridization is sp . If $X = 3$, it means three hybrid orbitals are to be formed. Hence, hybridisation is sp^2 and so on, as given in the following table :

Value of X	2	3	4	5	6	7
Type of hybridisation	sp	sp^2	sp^3	sp^3d	sp^3d^2	sp^3d^3

10. VALENCE BOND THEORY

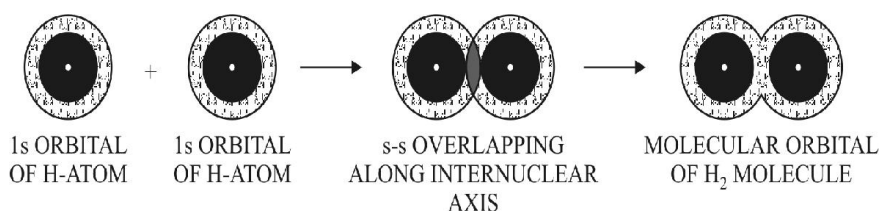
10.1 Types of covalent bond

Depending upon the type of overlapping, the covalent bonds are mainly of two types :

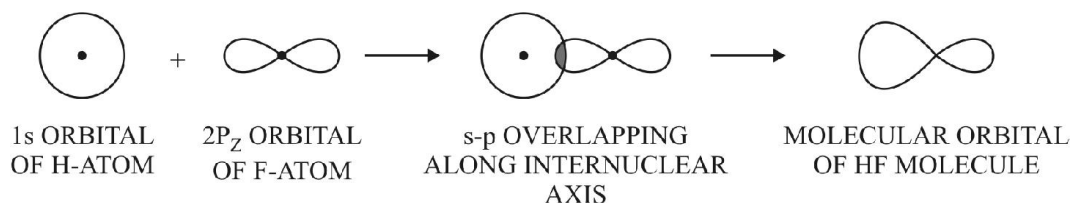
1. Sigma (σ) bond

When a bond is formed between two atoms by the overlap of their atomic orbitals along the internuclear axis (end to end or head on overlap), the bond formed is called sigma (σ) bond.

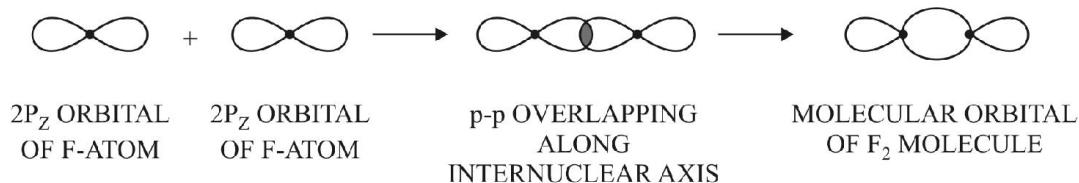
(i) s-s overlapping



(ii) s-p overlapping

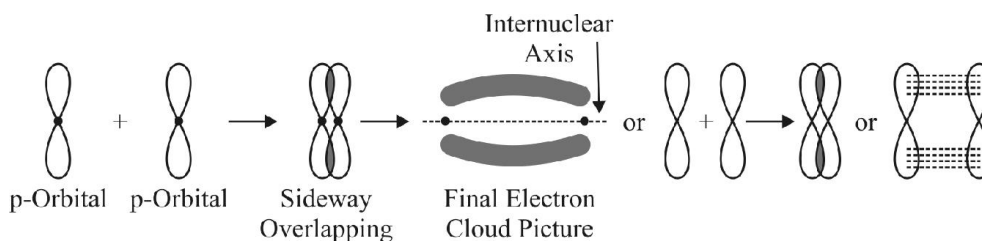


(iii) p-p overlapping



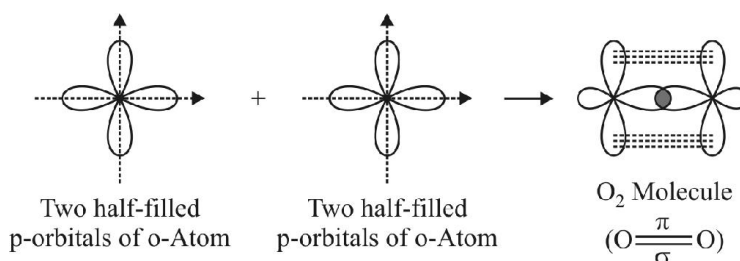
2. Pi (π) Bond

Pi-bond is formed by lateral (sideways) overlapping of p-orbitals, i.e., by overlapping of p-orbitals in a direction at right angles to the internuclear axis (figure).



p-p overlapping forming a pi bond

- (i) In case of oxygen molecule (each oxygen atom having electronic configuration, 1s² 2s² 2p_x² 2p_y¹ 2p_z¹), the two atoms are held together by one σ-bond and one π-bond as shown in figure.



Formation of oxygen molecule

Table : Comparison and sigma and pi bonds

Sigma (σ) Bond	Pi (π) Bond
(i) This bond is formed by overlap of orbitals along their internuclear axis (end to end overlap).	(i) This is formed by sideways overlapping of orbitals (lateral overlapping).
(ii) This is formed by overlapping between s-s, s-p or p-p orbitals.	(ii) This is formed by the overlap of p-p orbitals only.
(iii) Overlapping is quite large and hence sigma bond is a strong bond.	(iii) Overlapping is to a small extent. Hence, π -bond is a weak bond.
(iv) Electron cloud in this case is symmetrical about the line joining the two nuclei.	(iv) Electron cloud of π -bond is unsymmetrical.
(v) Sigma bond consists of only one electron cloud, symmetrical about the internuclear axis.	(v) Pi (π) bond consists of two electron clouds, one above the plane of atomic nuclei and the other below it.
(vi) Free rotation about a σ -bond is possible.	(vi) Free rotation about a π -bond is not possible because on rotation, overlapping vanishes and so the bond breaks.

10.2 Bond Parameters

10.2.1 Bond length

The equilibrium distance between the centres of the nuclei of the two bonded atoms is called its bond length.

10.2.1.1 Factors affecting bond length

- Size of the atoms : The bond length increases with increase in the size of the atoms. For example, bond lengths of H-X are in the order :
 $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
- Multiplicity of bond : The bond length decreases with the multiplicity of the bond. Thus, bond length of carbon-carbon bonds are in the order :
 $\text{C} \equiv \text{C} < \text{C} = \text{C} < \text{C} - \text{C}$
- Type of hybridisation (discussed later in sec. 4.24). As an s-orbital is smaller in size, greater the s-character, shorter is the hybrid orbital and hence shorter is the bond length. For example,
Bond lengths : $\text{sp}^3 \text{C-H} > \text{sp}^2 \text{C-H} > \text{sp} \text{C-H}$
s-character : (25%) (33%) (50%)

10.2.2 Bond energy

The amount of energy required to break one mole of bonds of a particular type so as to separate them into gaseous atoms is called bond dissociation enthalpy or simply bond enthalpy.

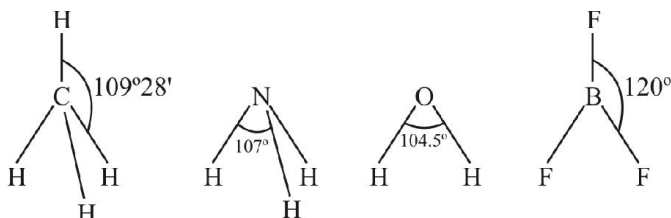
10.2.2.1 Factors affecting Bond energy

- Size of the atoms : Greater the size of the atoms, greater is the bond length and less is the bond dissociation enthalpy, i.e., less is the bond strength.
- Multiplicity of bonds : For the bond between the same two atoms, greater is the multiplicity of the bond, greater is the bond dissociation enthalpy. This is firstly because atoms come closer and secondly, the number of bonds to be broken is more. For example, bond dissociation enthalpies of H_2 , O_2 and N_2 are in the order :
 $\text{H-H} < \text{O=O} < \text{N} \equiv \text{N}$
- Number of lone pairs of electrons present : Greater the number of lone pairs of electrons present on the bonded atoms greater is the repulsion between the atoms and hence less is the bond dissociation enthalpy. For example, for a few single bonds, we have

Bond	C-C (in $\text{H}_3\text{C}-\text{CH}_3$)	N-N (in $\text{:N} \equiv \text{N:}$)	O-O (in $\text{:}\ddot{\text{O}}-\ddot{\text{O}}\text{:}$)	F-F ($\text{:}\ddot{\text{F}}-\ddot{\text{F}}\text{:}$)
Lone pairs of electrons on each atom	0	1	2	3
Bond enthalpy (kJ mol^{-1})	348	163	146	139

10.2.3 Bond angle

The angle between the lines representing the directions of the bonds, i.e., the orbitals containing the bonding electrons is called the bond angle.



10.2.4 Bond order

In the Lewis representation of a molecule or ion, the number of bonds present between two atoms is called bond order. For example, the bond orders of a few molecules are given below :

Molecule :	$\text{H}-\text{H}$	$\text{O}=\text{O}$	$\text{N}\equiv\text{N}$	$\text{C}\equiv\text{O}$
Bond order :	1	2	3	3

For odd electron molecules, as the three electron bond is considered as equivalent to half covalent bond, bond order can be fractional also. For example, Lewis structure of NO is



11. MOLECULAR ORBITAL THEORY

There is another approach to chemical bonding known as molecular orbital theory (MOT) developed by **Mulliken** (1932) and **Hund**, which explains the bonding characteristics in a better way. The molecular orbital theory considers the entire molecule as a unit with all the electrons moving under the influence of all the nuclei present in the molecular. This approach recognizes that each electron belongs to the molecule as a whole and may move within the entire molecule.

11.1 Molecular Orbitals

When the atoms to be bonded come close together, the orbitals of the bonded atoms lose their individual character and fuse (overlap) to form larger orbitals called molecular orbitals. Like atomic orbitals, there are molecular orbitals in a molecule. The only difference is that in atomic orbitals, electrons move under the influence of only one nucleus (i.e. Atomic orbital are monocentric), while in molecular orbitals, electrons move under the influence of many nuclei, they are polycentric.

11.2 Important features of M.O.T.

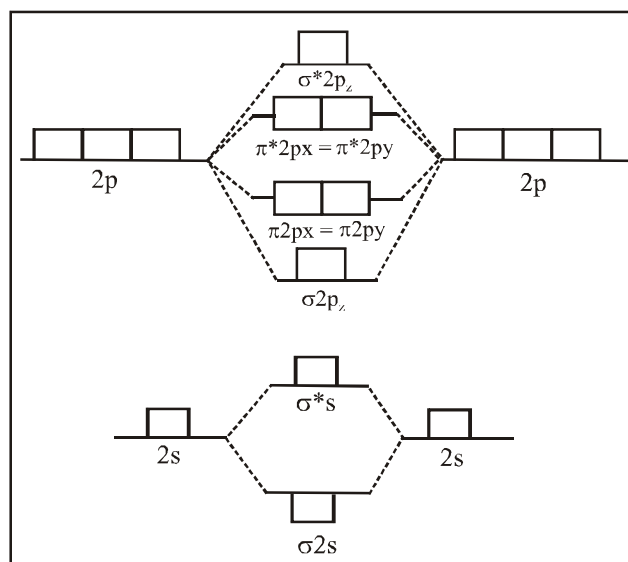
- Like an Atomic orbital which is around the nucleus of an atom there are molecular orbital which are around the nuclei of a molecule.
- The molecular orbitals are entirely different from the atomic orbitals from which they are formed.
- The valence electrons of the constituent atoms are considered to be moving under the influence of nuclei of participating atoms in the molecular orbital.
- The molecular orbitals possess different energy levels like atomic orbitals in an isolated atom.
- The shape of molecular orbitals are dependent upon the shapes of atomic orbitals from which they are formed.
- Molecular orbitals are arranged in order of increasing energy just like atomic orbitals.
- The number of molecular orbitals formed is equal to the number of atomic orbitals combining in bond formation.
- Like atomic orbitals, the filling of electrons in molecular orbitals is governed by the three principles such as **Aufbau principle**, **Hund's rule** and **Pauli's exclusion principle**.

11.3 Conditions for atomic orbitals to form M.O.

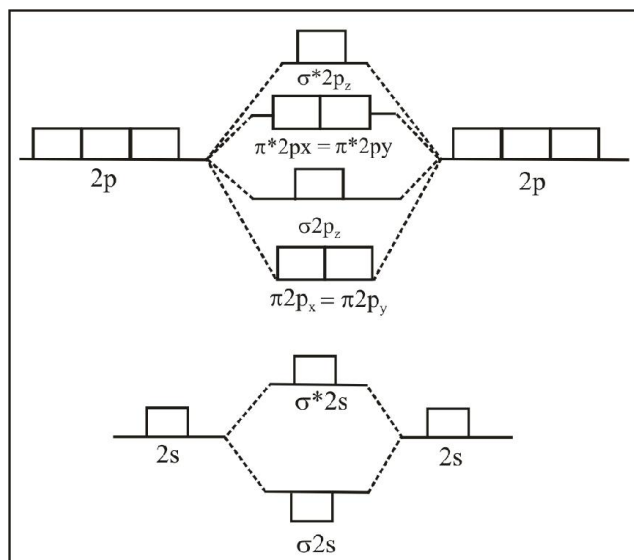
- The combining A.O. should be of a comparable energy.
- The combining atomic orbitals must overlap to a large extent greater the overlap, stable is the molecule formed.

11.4 Relative energies of M.O. and filling of electron

Energy diagram is shown below



M.O Energy level diagram for O_2 , F_2 and Ne.

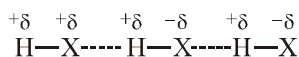
M.O energy diagram for Li_2 , Be_2 , B_2 , C_2 and N_2 molecule

12. METALLIC BOND

The constituent particles of metallic solids are metal atoms which are held together by metallic bond. A metal atom is supposed to consist of two parts, valence electrons and the remaining part (the nucleus and the inner shells) called **kernel**. The kernels of metal atoms occupy the lattice sites while the space between the kernel is occupied by valence electrons. Due to small ionisation energy the valence electrons of metal atoms are not held by the nucleus firmly. Therefore, the electrons leave the field of influence of one kernel and come under the influence of another kernel. Thus the electrons are not localised but are mobile. The simultaneous attraction between the kernels and the mobile electrons which hold the kernel together is known as **metallic bond**. This model is known as electron sea model.

13. HYDROGEN BONDING

When a hydrogen atom linked to a highly electronegative atom (like F, O, or N) comes under the influence of another strongly electronegative atom, then a weak bond is developed between them which is called as **hydrogen bond**. It is represented by dotted line as follows



As a result of hydrogen bonding, a H-atom links the two electronegative atoms simultaneously, one by a covalent bond and the other by a hydrogen bond. Hence it is said to form a hydrogen bridge. It is merely a strong electrostatic attractive forces and not a normal chemical bond. It is very weak (strength about 2-10 K cal/mol).

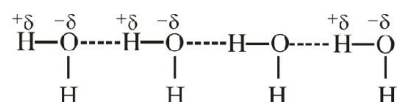
13.1 Conditions for Hydrogen bonding

- The molecule must contain a highly electronegative atom linked to H-atom. The higher the electronegativity, more is the polarisation of the molecule.
- The size of the electronegative atom should be small. The smaller the size the greater is the electrostatic attraction.

13.2 Types of Hydrogen bonding

13.2.1 Intermolecular Hydrogen bonding

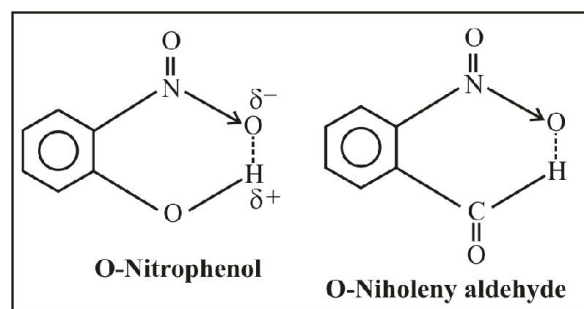
When hydrogen bonding takes place between different molecules of the same or different compounds, it is called intermolecular hydrogen bonding e.g. HF, H_2O , ROH (same compound) water-alcohol, water ammonia (different compound) etc.



13.2.2 Intramolecular hydrogen bonding

The hydrogen bonding which takes place within a molecule itself. It takes place in compounds containing two groups such that one group contain a H-atom linked to an electronegative atom and the other group contains a highly electronegative atom linked to a lesser electronegative atom. The bond is formed between the H-atom of one group with the more electronegative atom of the other group.

Examples are as shown below



14. VANDER WAAL'S FORCES

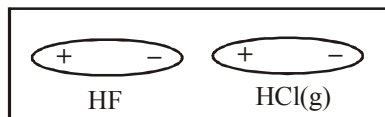
- This type of attractive forces occurs in case of non polar molecules such as H_2 , O_2 , Cl_2 , CH_4 , CO_2 , etc.
- The existence of weak attractive forces among the nonpolar molecule was first proposed by Dutch scientist **J.D. Vander Waal**.
- Vander waal force $\propto$ molecular weight
 $\propto$ Boiling point

14.1 Types of Vander Waal's force

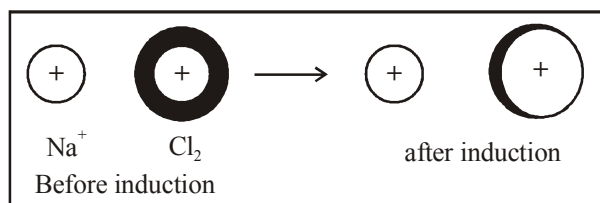
14.1.1 Ion dipole attraction - This force is between an ion such as Na^+ and a polar molecule such as HCl



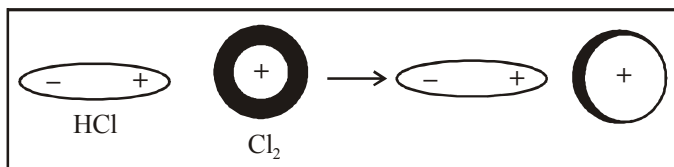
14.1.2 Dipole dipole attraction - It is again in between two polar molecules such as HF and HCl



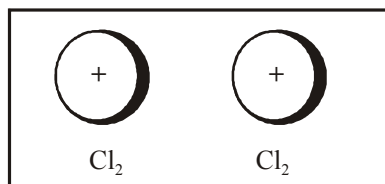
14.1.3 Ion induced dipole attraction - In this case a neutral molecule is induced by an ion as a dipole as shown in fig



14.1.4 Dipole - induced dipole attraction : In this case a neutral molecule is induced as a dipole by another dipole as shown in fig



14.1.5 Induced dipole - induced dipole attraction or London dispersion force between two non polar molecules as in Cl_2 , He etc.

**NOTE**

The relative strength of various bonds is as follows :
Ionic bond > Covalent bond > Metallic bond > H-bond
> Vander waal bond.

15. DIPOLE MOMENT

A molecule with positive and negative charge centres in equilibrium is called as dipole and is characterised by possessing a quantity dipole moment (μ) defined as the product of the magnitude of charge (q) and the distance (d) separating the centres of +ve and -ve charges. Its direction is from +ve end to -ve end.

$$\mu = q \times d$$

The charge q on an electron is 4.8×10^{-10} esu. The distance is measured in cm and is of order of $1 \text{ \AA}$, i.e., 10^{-8} cm. Its unit in CGS system is debye (D).

A molecule will have a dipole moment of 1 Debye (D) if charges of 1×10^{-8} esu are separated by a distance of $1 \text{ \AA}$.

Thus, $1\text{D} = 1 \times 10^{-18}$ esu. cm.

Applications of Dipole Moment

- The molecules having zero moment are non-polar molecules and those having $\mu_{\text{net}} \neq 0$ are polar in nature.
- The value of dipole moment can be used for determining the amount of ionic character in a bond.

Percentage of ionic character

$$= \frac{\text{Experimental value of dipole moment}}{\text{Theoretical value of dipole moment}} \times 100$$

SOLVED EXAMPLES

Example - 1

Which of the following compounds has the largest dipole moment ?

- (i) CH_3OH (ii) CH_4
(iii) CF_4 (iv) CO_2
(v) CH_3F

Sol. CH_4 and CF_4 have tetrahedral structure and are symmetrical, hence their dipole moment is zero, CO_2 is linear and hence its dipole moment is also zero of the remaining CH_3OH and CH_3F , since F is more electronegative than O, CH_3F will have high dipole moment.

Example - 2

What type of bond is formed when atoms have

- (i) Zero difference of electronegativity.
(ii) Little difference of electronegativity.
(iii) High difference of electronegativity.

Sol. (i) Non-polar covalent
(ii) Polar covalent
(iii) Electro-valent.

Example - 3

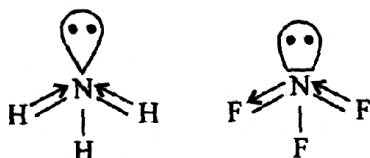
Explain the formation of a chemical bond.

Sol. A chemical bond is formed by mutual sharing or by transfer of one or more electrons. In doing so, each combining atom acquires stable noble gas electronic configuration having 8 electrons in its outermost shell.

Example - 4

Why NH_3 has higher dipole moment than NF_3 ?

Sol. In NH_3 dipoles are towards lone pair whereas in NF_3 dipoles are opposite to lone pair of electrons.

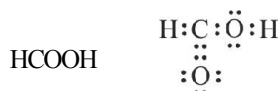
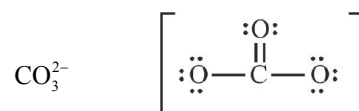
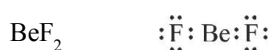
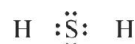


Example - 5

Draw the Lewis structures for the following molecules and ions :

H_2S , SiCl_4 , BeF_2 , CO_3^{2-} , HCOOH .

Ans. H_2S



Example - 6

Write the favorable factors for the formation of ionic bond.

Sol. Factors which favour the formation of ionic bond :

- Low ionisation energy** : In the formation of ionic bond a metal atom loses electron to form cation. This process requires energy equal to the ionisation energy. Lesser the value of ionisation, greater will be the tendency of the atom to form cation.
- High electron affinity** : The value of electron affinity gives the tendency of an atom to form anion. Greater the value of electron affinity, more will be the tendency of an atom to form anion.
- High lattice energy** : The lattice energy should be high so that it overcomes the apparent deficit of energy of absorption, that is, sublimation energy, ionization energy, dissociation energy and second affinity. Higher the lattice energy, greater will be the ease of formation of the ionic compound.

The magnitude of lattice energy gives an idea about the interionic forces. It depends upon the following factors :

- Size of the ions** : Smaller the size of the ions, lesser is the inter-nuclear distance. Consequently, the interionic attractions will be high and the lattice energy will also be large.

- (b) **Charge on the ions:** Larger the magnitude of charge on the ions, greater will be the attractive forces between the ions. Consequently, the lattice energy will be high.

Example - 7

Discuss the shape of the following molecules using the VSEPR model.

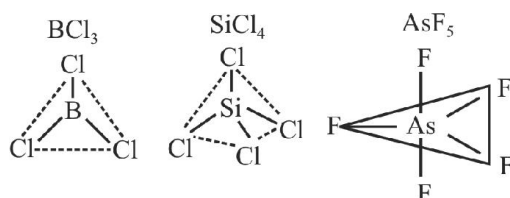
BeCl_2 , BCl_3 , SiCl_4 , AsF_5 , H_2S , PH_3 .

Sol. BeCl_2 : A linear molecule. Be atom has 2 electrons in its outermost orbit. Each chlorine atom has seven valence electrons. The Lewis structure of BeCl_2 is



There are two electron pairs and to minimise the repulsion, these electron pairs tend to keep themselves far away from each other, i.e., 180° apart. This gives BeCl_2 a linear structure.

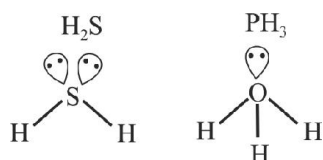
BCl_3 : In BCl_3 molecule, the three bond pairs of electrons are located around B in a triangular arrangement. Thus, the molecule BCl_3 has a **triangular planar geometry**.



SiCl_4 : A tetrahedral molecule. Si has 4 electrons in its outermost shell. Due to mutual sharing of electrons with Cl there are 4 electron pairs around Si. To keep the repulsion at the minimum, these 4 electron pairs should be arranged in a tetrahedral manner around Si. Thus, SiCl_4 is a tetrahedral molecule.

AsF_5 : Trigonal bipyramidal molecule : As has five electrons in its outermost orbit. Due to sharing of 5 electrons from 5 F-atoms, there are in all 5 electron pairs. These are distributed in space to form a trigonal bipyramid.

H_2S : Bent (V-shaped) structure : S has 6 electrons in its outermost shell. 2 H-atoms contribute 2 electrons during bonding. Thus, there are 8 electrons or 4 electron pairs around S. This gives a tetrahedral distribution of electron pairs around S. The two corners of the tetrahedron are occupied by H-atoms and the other two by the lone-pairs of electrons. Thus, H_2S has a bent structure.

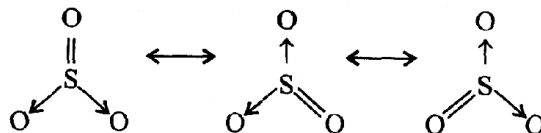


PH_3 : Trigonal pyramidal : Phosphorus atom has 5 electrons in its outermost orbit. H-atoms contribute one electron each to make in all 8 electrons around P-atom. Thus, 4 pairs of electrons would be distributed in a tetrahedral fashion around the central atom. Three pairs form three P-H bonds while the fourth pair remains unused. Due to repulsion between the bonding and lone pairs of electrons, the angle $\hat{\text{H}}\text{P}\text{H}$ is not exactly tetrahedral ($109^\circ 28'$). The actual HPH angle is 93.3° . Thus PH_3 is a trigonal pyramidal molecule.

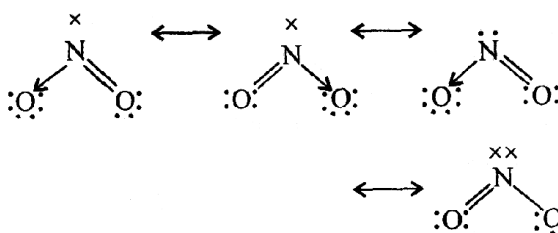
Example - 8

Write the resonance structures for SO_3 , NO_2 and NO_3^- .

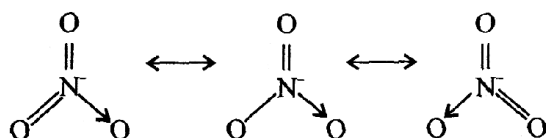
Sol. (SO_3)



(NO_2)



(NO_3^-)

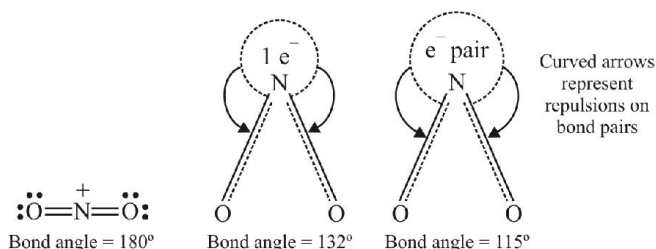


Example - 9

Arrange the following in order of decreasing bond angle giving reason

NO_2 , NO_2^+ , NO_2^-

Ans. $\text{NO}_2^+ > \text{NO}_2 > \text{NO}_2^-$. This is because NO_2^+ has no lone pair of electrons (i.e. has only bond pairs on two sides) and hence it is linear. NO_2 has one unshared electron while NO_2^- has one unshared electron pair. There are greater repulsions on N-O bonds in case of NO_2^- than in case of NO_2 .



Example - 10

Arrange the following in order of decreasing bond angles

(i) CH_4 , NH_3 , H_2O , BF_3 , C_2H_2 (ii) NH_3 , NH_2^- , NH_4^+

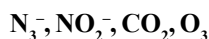
Sol. (i) $\text{C}_2\text{H}_2 (180^\circ) > \text{CH}_4 (109^\circ 28') > \text{BF}_3 (120^\circ) > \text{NH}_3 (107^\circ) > \text{H}_2\text{O} > (104.5^\circ)$

(ii) $\text{NH}_4^+ > \text{NH}_3 > \text{NH}_2^-$

This is because all of them involve sp^3 hybridization. The number of lone pair of electrons present on N-atom are 0, 1 and 2 respectively. Greater the number of lone pairs, greater are the repulsions on the bond pairs and hence smaller is the angle.

Example - 11

Which of the following species have same shape/same bond order?



Sol. Isoelectronic species have same shape/same bond order. The number of outer electrons is same in CO_2 and N_3^- and same in O_3 and NO_2^- .

Example - 12

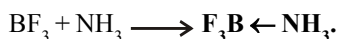
Although both CO_2 and H_2O are triatomic molecules, the shape of H_2O molecule is bent while that of CO_2 is linear, Explain this on the basis of dipole moment.

Sol. The H_2O molecule has a dipole moment of 1.84 D. There are two OH bonds in H_2O molecule. The O–H bonds are polar and has dipole moment of 1.5 D. Since the water molecule has a net dipole, hence the two O–H dipoles are not in a straight line opposing each other. This rules out the linear structure (H–O–H) for water. The two O–H bonds cannot lie along the same line in the same direction also. Therefore, the two O–H bonds in H_2O molecule must be inclined to each other at certain angle. Thus, H_2O molecule has an angular shape.

The CO_2 molecule has zero dipole moment. In CO_2 , there are two C = O bonds. Each C = O bond is a polar bond. This means each bond has a dipole moment. Since, the net dipole, and hence the two bonds must be just opposite to each other, i.e., the two bonds must be at 180° relative to each other. Thus CO_2 is a linear molecule.

Example - 13

Is there any change in the hybridization of B and N atoms as a result of the following reaction?



Sol. Here, B atom in BF_3 is sp^2 hybridized and one of its p orbital is empty. N in NH_3 is sp^3 hybridized and one of its hybrid orbitals is occupied by a lone-pair of electrons. During the reaction a coordinate bond is formed due to one-side sharing of electron pair.

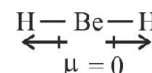


There is no change in the hybridization of any of the two atoms in this reaction.

Example - 14

Explain why BeH_2 molecule has a zero dipole moment although the Be–H bonds are polar.

Sol. This is because BeH_2 is a linear molecule and the two bond dipoles oriented at an angle of 180° neutralize each other. As a result, the net dipole moment of the molecule is zero.



Example - 15

Describe the change in hybridization (if any) of the Al atom in the following reaction.



Sol. $\text{AlCl}_3 + \text{Cl}^- \rightarrow \text{AlCl}_4^-$
tetrahedral geometry
 sp^3 hybridization

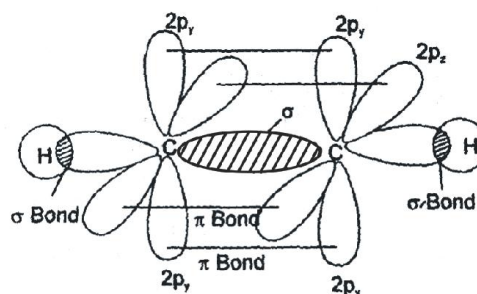
Aluminium chloride exists as dimer Al_2Cl_6 . In this molecule chlorine atoms are arranged tetrahedrally around the two Al atoms.

Thus Al atoms show Sp^3 hybridisation in Al_2Cl_6 . Thus, there is no change in the hybridisation of Al atom in the given reaction.

Example - 16

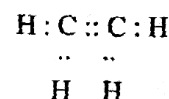
Draw diagrams showing the formation of a double bond and a triple bond between carbon atoms in C_2H_4 and C_2H_2 molecules.

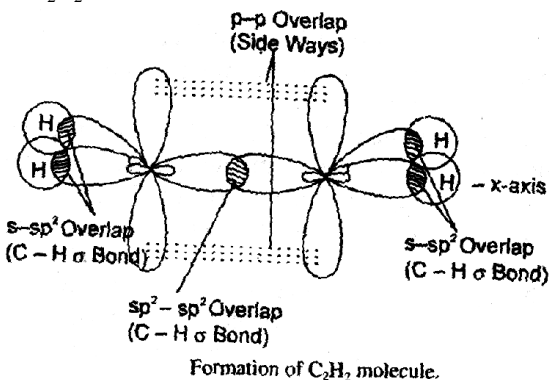
Sol. (i) C_2H_4



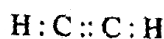
Formation of a double bond between carbon atoms in C_2H_4

The Lewis structure of ethylene is represented as



(ii) C_2H_2 

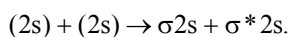
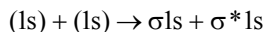
It is clear that the carbon-carbon triple bond is made up of one strong σ -bond and two weak π -bonds. The Lewis structure of acetylene can be represented as

**Example - 17**

Does Li_2 exist? If so, estimate its bond order and compare its bond dissociation energy with that of H_2 .

Sol. Li has a configuration : $1s^2, 2s^1$.

There are two s orbitals ($1s$ and $2s$) on each atom. These combine to give four MO's. These are



Thus all the six electrons are accommodated in these four orbitals. The electronic configuration for Li_2 is $(\sigma 1s)^2 (\sigma^* 1s)^2 (\sigma 2s)^2$.

$$N_b = 4, N_a = 2.$$

$$\text{Bond order in } Li_2 = \frac{4-2}{2} = 1.$$

Therefore, Li_2 should be a stable species. Its bond dissociation energy is 105 kJ mol^{-1} , as compared to 431 kJ mol^{-1} for H_2 . Thus, the bond in Li_2 is much weaker than that on H_2 . This is because $2s$ orbital of Li (involved in the bonding) is much larger than $1s$ orbital of H.

Example - 18

Explain how V.B theory differs from the Lewis concept.

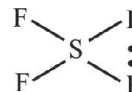
Sol. The Lewis concept of describes the formation of bond in terms of sharing of one or more electron pairs and the octet rule. It does not explain the energetics of the bond formation and shapes of the polyatomic molecules.

The VB theory describes the bond formation in terms of hybridization and overlap of the orbitals. The overlap of orbitals along the intermolecular axis increases the electron-density between the two nuclei resulting in a decrease in the energy and formation of a bond.

Example - 19

Why is it that in the SF_4 molecule, the lone pair of electrons occupies an equatorial position in the overall trigonal bi-pyramidal arrangement in preference to an axial position.

Sol. In SF_4 molecule, the lone pair of electrons occupies an equatorial position because in this geometry (sp^3d hybridisation), the lone pair-bond pair repulsion is minimum

**Example - 20**

Define hydrogen bond. Is it weaker or stronger than the Van der Waals forces?

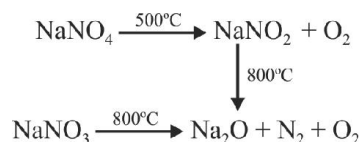
Sol. The hydrogen bond can be defined as the attractive force which binds hydrogen atom of one molecule with the electronegative atom (F, O or N) of another molecule. A hydrogen bond is shown by a dotted line (.....)

Hydrogen bond is weaker than the Van der Waals forces.

Example - 21

$NaNO_3$ on heating to $500^\circ C$ decomposes to produce $NaNO_2$ and O_2 , while the same salt on heating to $800^\circ C$, decomposes to Na_2O , N_2 and O_2 . Explain in light of Fajan's rule.

Sol. The decomposition scheme can be shown as



As we know the thermal stability of a salt depends on percentage ionic character, $NaNO_2$ has greater ionic character than $NaNO_3$ because of lower polarizability of smaller nitrite ion than nitrate ion. Therefore, $500^\circ C$ is sufficient for decomposition of less ionic $NaNO_3$ but $NaNO_2$ is stable at this temperature. On heating further to $800^\circ C$, the more stable $NaNO_2$ decomposes further to Na_2O and N_2, O_2 .

Example - 22

Out of σ and π – bonds, which one is stronger and why ?

Sol. σ -bond is stronger. This is because σ -bond is formed by head-on overlapping of atomic orbitals and, therefore, the overlapping is large. π -bond is formed by sideways overlapping which is small.

Example - 23

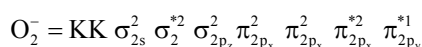
Out of p-orbital and sp-hybrid orbital which has greater directional character and why ?

Sol. sp-orbital has greater directional character than p-orbital. This is because p-orbital has equal sized lobes with equal electron density in both the lobes whereas sp-hybrid orbital has greater electron density on one side.

Example - 24

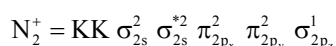
Compare the relative stabilities of O_2^- and N_2^+ and comment on their magnetic (paramagnetic or diamagnetic) behaviour.

Sol. M.O. Electronic configuration of



$$\text{Bond order} = \frac{1}{2} (8 - 5) = \frac{3}{2} = 1.5$$

M.O. Electronic configuration of



$$\text{Bond order} = \frac{1}{2} (7 - 2) = \frac{5}{2} = 2.5$$

As bond order of N_2^+ > bond order of O_2^- , therefore, N_2^+ is more stable than O_2^- .

Each of them contains unpaired electron, hence both are paramagnetic.

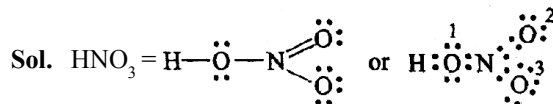
Example - 25

Explain why HF is less viscous than H_2O .

Sol. There is greater intermolecular hydrogen bonding in H_2O than that in HF as each H_2O molecule forms four H-bonds with other water molecules whereas HF forms only two H-bonds with other HF molecules. Greater the intermolecular H-bonding, greater is the viscosity. Hence, HF is less viscous than H_2O .

Example - 26

Write Lewis structure of the following compounds and show formal charge on each atom :



$$\text{Formal charge} = V - L - \frac{1}{2} S$$

$$\text{Formal charge on H} = 1 - 0 - \frac{1}{2} \times 2 = 0$$

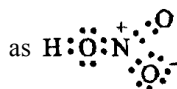
$$\text{Formal charge on O (1)} = 6 - 4 - \frac{1}{2}(4) = 0$$

$$\text{Formal charge on N} = 5 - 0 - \frac{1}{2}(8) = 1$$

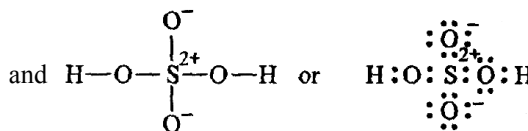
$$\text{Formal charge on O (2)} = 6 - 4 - \frac{1}{2}(4) = 0$$

$$\text{Formal charge on O (3)} = 6 - 6 - \frac{1}{2}(2) = -1$$

Hence, we write the structure of HNO_2 with formal charges



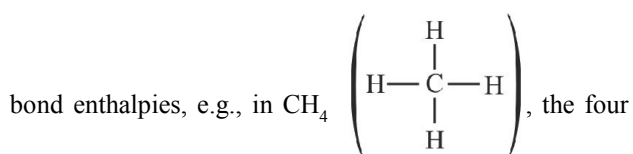
Similarly, we have  or 



Example - 27

What is mean by the term average bond enthalpy ? Why is there difference in bond enthalpy of O-H bond in ethanol ($\text{C}_2\text{H}_5\text{OH}$) and water ?

Sol. All the similar bonds in a molecule do not have the same



C-H bonds do not have the same bond enthalpies because after breaking of the bonds one by one, the electronic environment around carbon changes. Hence, the actual bond enthalpy of C-H bond is taken as the average value.

O-H bond in ethanol ($\text{C}_2\text{H}_5-\text{O}-\text{H}$) and that in water

($\text{H}-\text{O}-\text{H}$) do not have similar electronic environment around oxygen atom. Hence, their O-H bond enthalpies are different.

Example - 28

Write Lewis dot structures of the following molecules/ions :

- (i) CO (ii) HCN

Sol : V = number of valence electron of molecule

R = Number of electron required for octet in molecule

S = Number of shared electron ($S = R - V$)

U = Number of unshared electron ($U = V - S$)

- (i) CO

$$V = 4 + 6 = 10 \text{ electrons}$$

$$R = 8 + 8 = 16 \text{ electrons}$$

$$S = R - V = 6 \text{ electrons}$$

$$U = 10 - 6 = 4 \text{ electrons}$$



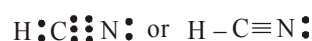
- (ii) HCN

$$V = 1 + 4 + 5 = 10 \text{ electrons}$$

$$R = 2 + 8 + 8 = 18 \text{ electrons}$$

$$S = 8 \text{ electrons}$$

$$U = 2 \text{ electrons}$$

**Example - 29**

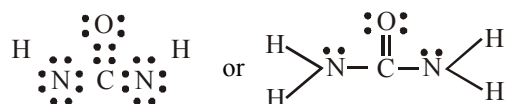
Write Lewis structures for

- (i) Urea, NH_2CONH_2

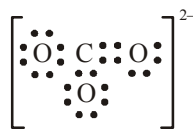
- (ii) Carbonate ion, CO_3^{2-} .

Sol : (i) Urea, V = 24 electrons, R = 40 electrons, S = 16 electrons

$$U = 8 \text{ electrons}$$



- (ii) V = 24 electrons, R = 32 electrons, S = 8 electrons, U = 16 electrons.

**Example - 30**

Arrange the following in increasing order of bond angle around the central atom :



Sol. $\text{H}_2\text{O} < \text{NH}_3 < \text{CH}_4 < \text{BF}_3 < \text{BeF}_2$

Bond angle 104.5° 106.5° $109^\circ 28'$ 120° 180°

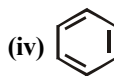
Example - 31

Indicate the number of σ and π -bonds in the following molecules.

- (i) $\text{CH}_3-\text{CH}=\text{CH}_2$

- (ii) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_3$

- (iii) $\text{CH}_3-\text{C} \equiv \text{C}-\text{CH}_3$



Sol. For finding out the number of σ and π -bonds in a molecule, the following points should be kept in mind :

1. All single bonds are sigma bonds.
 2. All double bonds consist of one σ and two π -bonds.
 3. All triple bonds consist of one σ and two π -bonds.
- (i) This molecule consists of seven single bonds and one double bond. Thus, number of σ -bonds = 8 and number of π -bonds = 1.
- (ii) This molecule consists of 13 single bonds. Thus, number of σ -bonds = 13.
- (iii) This molecule consists of 8 single bonds and one triple bond. Thus, number of σ -bonds = 9 and number of π -bonds = 2.
- (iv) The molecule of benzene consists of nine single bonds and three double bonds. Thus, number of σ -bonds = 12 and number of π -bonds = 3.

Example - 32

Which of the two, peroxide ion or superoxide ion, has a larger bond length ?

Sol. The bond length in a molecule depends on bond order. the higher the bond order, smaller will be the bond length.

Peroxide ion, O_2^{2-}

Configuration KK $\sigma(2s)^2 \sigma^*(2s)^2 \sigma(2p_z)^2 \pi(2p_x)^2$

$\pi(2p_y)^2 \pi^*(2p_x)^2 \pi^*(2p_y)^2$

$$\text{Bond order} = \frac{8-6}{2} = 1$$

Superoxide ion, O_2^-

Configuration KK $\sigma(2s)^2 \sigma^*(2s)^2 \sigma(2p_z)^2 \pi(2p_x)^2$

$\pi(2p_y)^2 \pi^*(2p_x)^2 \pi^*(2p_y)^1$

$$\text{Bond order} = \frac{8-5}{2} = 1.5$$

Bond order of superoxide is higher than peroxide ion, hence bond length of peroxide ion is larger.

Example - 33

Which of the following has maximum bond angle ?

H_2O, CO_2, NH_3, CH_4

Sol. CO_2 (Linear structure, carbon undergoes sp -hybridization).

Example - 34

Interpret the non-linear shape of H_2S molecule and non planar shape of PCl_3 using valence shell electron pair repulsion (VSEPR) theory.

Sol. In H_2S , two bonded pairs and two lone pairs of electrons are present, i.e., S undergoes sp^3 -hybridization. Tetrahedral configuration comes into existence. Two of the positions are occupied by lone pairs, hence the actual structure is bent or V-shaped.

In PCl_3 , three bonded pairs and one lone pair of electrons are present, i.e., P-atom undergoes sp^3 - hybridization. Tetrahedral configuration comes into existence. One position is occupied by lone pair, hence the actual structure is pyramidal.

Example - 35

Hydrogen bonding does not exist in HCl through chlorine is quite electronegative.

Sol. This is due to the large size of Cl-atom

Example - 36

The dipole moment of KCl is 3.336×10^{-29} coulomb metre which indicates that it is a highly polarized molecule. The interatomic distance between K^+ and Cl^- in this molecule is 2.6×10^{-10} m. Calculate the dipole moment of KCl molecule, if there were opposite charges of one fundamental unit located at each nucleus. Calculate the percentage ionic character of KCl.

$$\mu = q \times d$$

Sol. q = one fundamental unit = 1.062×10^{-19} C

$$d = 2.6 \times 10^{-10} \text{ m}$$

$$= 1.602 \times 10^{-19} \times 2.6 \times 10^{-10}$$

$$= 4.1652 \times 10^{-29} \text{ coulomb metre}$$

Percentage ionic character

$$= \frac{3.336 \times 10^{-29}}{4.1652 \times 10^{-29}} \times 100 = 80.09$$

Example - 37

The correct order of the ionic character of PbF_2 , $PbCl_2$, $PbBr_2$ and PbI_2 is :

(a) $PbF_2 > PbCl_2 > PbBr_2 > PbI_2$

(b) $PbF_2 > PbBr_2 > PbCl_2 > PbI_2$

(c) $PbF_2 < PbCl_2 < PbBr_2 < PbI_2$

(d) $PbF_2 < PbCl_2 > PbBr_2 > PbI_2$

Sol. Since the given dihalides contain the same cation viz Pb^{2+} but different anions (halide ions), the ionic character of these halides depends on the size of the halide ions. The ionic character decreases with the increases in the size of the halide ions. Thus since the size of halides ions increases as $F^- < Cl^- < Br^- < I^-$, the ionic character of PbX_2 molecules decreases as $PbF_2 > PbCl_2 > PbBr_2 > PbI_2$. Thus (a) is the correct answer.

Example - 38

Among the halides namely, NaF, NaCl, NaBr and NaI, NaF has the highest melting point, because NaF has :

(a) minimum ionic character

(b) maximum ionic character

(c) highest oxidising power

(d) lowest valency

Sol. Since the size of halide ions (X^-) increases as $F^- < Cl^- < Br^- < I^-$ the ionic character of MX molecules decreases as $NaF > NaBr > NaI$. Being the most ionic, NaF has the highest melting point.

Example - 39

A diatomic molecule has a dipole moment equal to 1.2 D. If its bond distance is 1.0 Å, what fraction of an electronic charge exists in each atom ?

- (a) 12% (b) 18%
(c) 25% (d) 29%

Sol. Bond distance, $d = 1.0 \text{ Å} = 1.0 \times 10^{-8} \text{ cm}$ (given)

Dipole moment, $\mu = 1.2 \text{ D} = 1.2 \times 10^{-18} \text{ esu. cm}$ (given)

Electronic charge = $4.8 \times 10^{-18} \text{ esu. cm}$ (given)

Fraction of electronic charge = x (to be calculated)

Now we know that :

$$\mu = x \times d$$

$$1.2 \times 10^{-18} \text{ esu. cm} = x \times 10^{-8} \text{ cm}$$

$$\therefore x = \frac{1.2 \times 10^{-18} \text{ esu. cm}}{10^{-8} \text{ cm}} = 1.2 \times 10^{-10} \text{ esu}$$

Out of $4.8 \times 10^{-10} \text{ esu. cm}$ electronic charge, the electronic charge existing on each atom = $1.2 \times 10^{-10} \text{ esu}$

$\therefore$ Out of 100 esu. cm. electronic charge, the electronic charge existing on each atom

$$= \frac{1.2 \times 10^{-10} \text{ esu}}{4.8 \times 10^{-10} \text{ esu. cm}} \times 100$$

$$= 25\%$$

Example - 40

Which of the following tetraiodides is the least stable and has doubtful existence ?

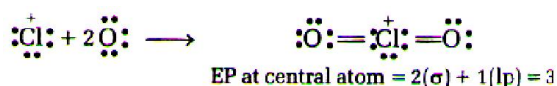
- (a) Cl_4 (b) GeI_4
(c) SnI_4 (d) PbI_4

Sol. The given tetraiodides contain C, Ge, Sn and Pb as C^{4+} , Sn^{4+} and Pb^{4+} ions respectively. According to inert pair effect, the stability of these cations decreases as $\text{C}^{4+} > \text{Ge}^{4+} > \text{Sn}^{4+} > \text{Pb}^{4+}$. This order shows that since Pb^{4+} ion is the least stable, its tetraiodide, PbI_4 is also least stable and hence has doubtful existence.

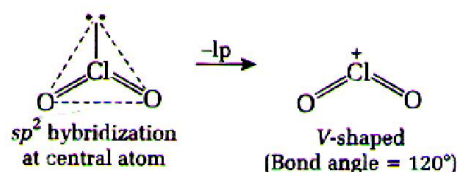
Example - 41

Discuss the bondings of ClO_2^+ ion.

Sol. The central atom is Cl^+ . The ions can be constituted as

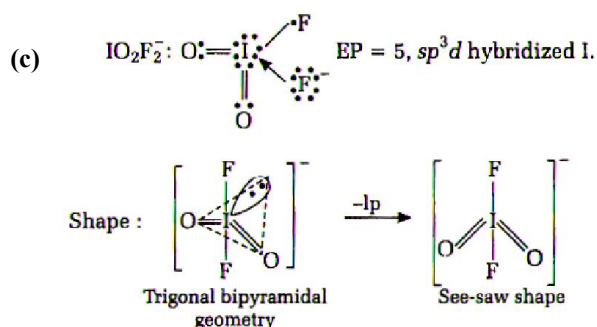
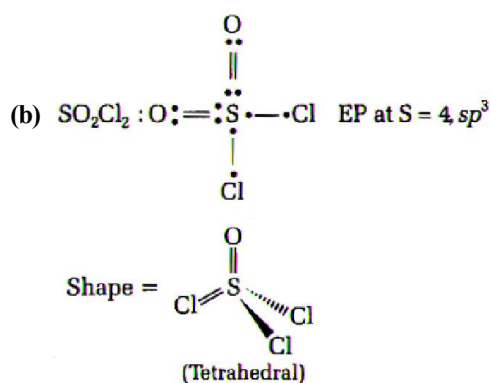
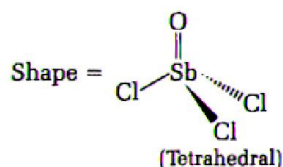
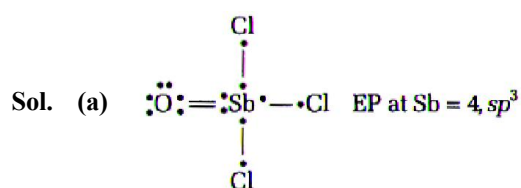


Geometry :


Example - 42

Write the Lewis dot formula of the following molecules and predict their molecular shapes :

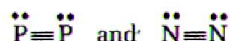
- (a) OSbCl_3 (b) SO_2Cl_2 (c) IO_2F_2^-



Example - 43

In vapour phase, phosphorus can exist as P_2 molecules, which is highly reactive, whereas N_2 is relatively inert, explain the difference.

Sol. Both P and N are from the same group (G15) of the Periodic Table, their diatomic molecule can be represented as :



As we know, Pi-bonding between two phosphorus atoms is very unlikely due to larger atomic radius. Hence, diatomic P_2 is unstable but there is not such restriction of Pi-bonding to N and other second period elements.

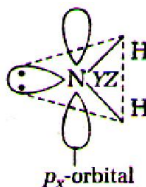
Example - 44

Both NH_2^- and NH_2^+ are angular species but the bond angle in NH_2^- is less than that in NH_2^+ . (a) What is the reason for this difference in bond angles? (b) Take the x-axis as lying perpendicular to the molecular plane, does N ($2p_x$) orbital participate in the hybridization of either of species?

Sol. (a) Hybridization at N atom is sp^3 in NH_2^- and sp^2 in NH_2^+ .

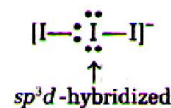
Therefore, bond angle in NH_2^- is close to 109° (tetrahedral angle), while bond angle in NH_2^+ is close to 120° (triangular planar angle).

(b) In NH_2^- , hybridization at N is $sp^3 \Rightarrow$ implies that all p-orbitals are involved in hybridization. In case of NH_2^+ , the hybridization at N is sp^2 and the ion lies in YZ plane as X-axis lying in perpendicular plane. For the species to be in YZ plane, the two p-orbitals involved in sp^2 hybridization must be p_y and p_z orbitals i.e., $2p_x$ orbitals at N is remaining unhybridized as

**Example - 45**

Although I_3^- is known, F_3^- is not. Explain.

Sol. In I_3^- , the central I is sp^3d hybridized and violate the octet rule. It is possible for elements of third and higher period to expand its valence shell beyond eight because they contain vacant d-orbitals. Hence formation of I_3^- is possible

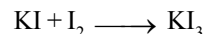


Since, fluorine is in second period, lack vacant d-orbitals, can't violate octet rule and, hence F_3^- does not exist.

Example - 46

Iodine (I_2) is a volatile solid at room temperature and has very low solubility in water. However, I_2 is highly soluble in an aqueous solution of KI. Explain.

Sol. I_2 (s) being a molecular solids, the molecules are completely non-polar and covalent, have very low affinity for polar solvents like water. However, in solution of KI, I_2 forms KI_3 salt as



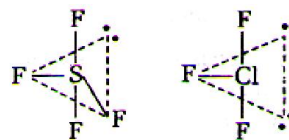
KI_3 , a strongly ionic salt dissolves easily in water like other normal salts of the alkali metals. Hence, I_2 becomes soluble in KI via KI_3 .

Example - 47

Compare the F (axial) — A — F (equatorial) bond angles in these molecules. PF_6 , SF_4 and ClF_3 .

Sol. PF_5 is symmetrical trigonal bipyramidal molecule hence the sought bond angle is 90° .

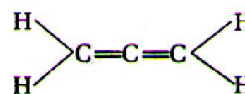
SF_4 is a see-saw shaped molecule, while ClF_3 is T-shaped as shown below :



Axial bonds in SF_4 is repelled by one lone pair, the same in ClF_3 are repelled by two lone pairs, more repulsions are observed in ClF_3 . Hence, F—S—F angle (axial-equatorial) is greater than F—Cl—F (axial-equatorial), although both are less than 90° . The order is : $PF_5 > SF_4 > ClF_3$.

Example - 48

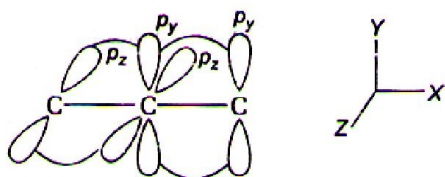
The Lewis structure of allene is



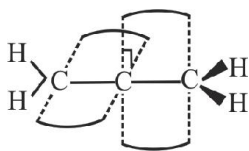
Make a three dimensional sketch of this molecule and answer these questions : (a) Is the molecule planar? (b) Does 1,3-dichloropropene show geometrical isomerism? (c) Is the molecule 1,3-dichloropropene polar?

Sol. Hybridization of central carbon is sp and two of its

unhybridized p-orbitals are involved in π -bonding. Also p-orbitals at an atom are orthogonal (at 90° angle), the two π -bonds at central carbon are also orthogonal. Let these p-orbitals, involved in π -bondings, at central carbon be p_y and p_z , then



Hybridization at the terminal carbons are both sp^2 . The two p-orbitals involved in sp^2 hybridization are p_x and p_y at left terminal and p_x and p_z at right terminals. Therefore, the three sp^2 hybrid orbitals at left terminal carbon are in xy plane, while the same at right terminal carbon are in xz plane. As a result, the two triangular planes at terminals are at right angles as shown below :



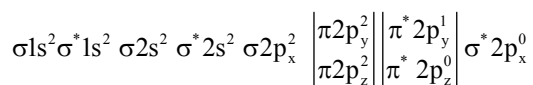
The two H—C—H planes at terminals are perpendicular to each other. (π -bonds are also perpendicular to each other.)

Hence, (a) Molecule is non-planar. (b) Doesn't show geometrical isomerism because two H at one terminal are equidistant from any H on other terminal. (c) Polar.

Example - 49

Order these species according to increasing C—F bond length : CF^+ , CF, CF^- . Predict if any is paramagnetic.

Sol. The molecular orbital electronic configuration of CF (15e) is



$$\text{Bond order} = \frac{10 - 5}{2} = 2.5$$

In CF^+ , there is one 'e' less in antibonding MO, therefore, bond-order = 3, while in CF^- , there is one more 'e' in antibonding MO giving bond-order = 2. Since 'Bond-order' is directly related to bond-energy, which is inversely proportional to bond-length, the order of bond-lengths in given series is :



Also, CF and CF^- are paramagnetic because of presence of one and two unpaired electrons respectively, while CF^+ has no unpaired electron and diamagnetic.

Example - 50

Both 'H' and 'F' are monovalent in covalent bonding, phosphorus forms PF_5 but no PH_5 exists. Explain.

Sol. Phosphorus has five electrons in the valence shell ($3s^2 3p^3$) as well as it contains vacant 3d orbitals, so there is scope for pentavalency. However, the 3d orbital of phosphorus is much larger in size than its 3s and 3p-orbitals, and do not usually hybridize. When highly electronegative atoms like F approach to P, its 3d-orbitals, and do not usually hybridize. When highly electronegative atoms like F approach to P, its 3d-orbitals hybridize with 3s and 3p orbitals and enable P to undergo sp^3d hybridization to form PF_5 , PCl_5 etc. Hydrogen being very less electronegative, cannot contract 3d orbitals of phosphorus to the level where it can participate in hybridization with 3s and 3p orbitals. Hence, with hydrogen, only s and p orbitals of P hybridize and no more than its three electrons can take part in sharing, showing trivalency only.

Example - 51

Trimethyl amine is a stronger base than trisilyl amine. Explain.

Sol. In trisilyl amine, a $p\pi-d\pi$ back bonding occurs as



The above $p\pi-d\pi$ bond decreases the availability of 1p at nitrogen, thus the basic strength. No such $p\pi-d\pi$ bonding is possible in trimethyl amine, hence stronger base.

EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Ionic Bonds

1. Which combination will give the strongest ionic bond ?
(a) K^+ and Cl^- (b) K^+ and O^{2-}
(c) Ca^{2+} and Cl^- (d) Ca^{2+} and O^{2-}

Bond Formation

2. Bond formation is
(a) always exothermic
(b) always endothermic
(c) neither exothermic nor endothermic
(d) sometimes exothermic and sometimes endothermic
3. Element X is strongly electropositive and element Y is strongly electronegative. Both are univalent. The compound formed would be
(a) X^+Y^- (b) X^-Y^+
(c) $X-Y$ (d) $X \rightarrow Y$
4. The bond between two identical non-metal atoms has a pair of electrons
(a) unequally shared between the two
(b) transferred fully from one atom to another
(c) with identical spins
(d) equally shared between them

Properties of Ionic cmpds

5. An ionic compound A^+B^- is most likely to be formed when:
(a) the ionization energy of A high and electron affinity of B is low
(b) the ionization energy of A is low and electron affinity of B is high
(c) both, the ionization energy of A and electron affinity of B are high
(d) both, the ionization energy of A and electron affinity of B are low
6. The lattice energy of KF, KCl, KBr and KI follow the order:
(a) $KF > KCl > KBr > KI$ (b) $KI > KBr > KCl > KF$
(c) $KF > KCl > KI > KBr$ (d) $KI > KBr > KF > KCl$
7. In which of the following species the bonds are non-directional ?
(a) NCl_3 (b) $RbCl$
(c) $BeCl_2$ (d) BCl_3

8. The compound with the highest degree of covalency is :

- (a) NaCl (b) $MgCl_2$
(c) AgCl (d) CsCl

Covalent Bond formation

9. The fluorine molecules is formed by :
(a) p-p orbitals (sideways overlap)
(b) p-p orbitals (end-to-end overlap)
(c) sp-sp orbitals (d) s-s orbitals
10. Covalency favoured in the following case :
(a) smaller cation (b) larger anion
(c) large charge on cation and anions
(d) all the above

Covalent bond properties

11. The molecular size of ICl and Br_2 is approximately same, but b.p. of ICl is about $40^\circ C$ higher than that of Br_2 . It is because:
(a) ICl bond is stronger than Br-Br bond
(b) IE of iodine < IE of bromine
(c) ICl is polar while Br_2 is nonpolar
(d) I has larger size than Br

Coordinate bond

12. Which of the following molecules does not have coordinate bonds ?
(a) PH_4^+ (b) NO_2
(c) O_3 (d) CO_3^{2-}

Covalent Bond

13. The compound which contains both ionic and covalent bonds is
(a) CH_4 (b) H_2
(c) KCN (d) KCl
14. The total number of electrons that taken part in forming the bonds in N_2 is
(a) 2 (b) 4
(c) 6 (d) 10

15. Which of the following compound is covalent ?
 (a) H_2 (b) CaO
 (c) KCl (d) Na_2S
16. Which pair of elements can form multiple bond with itself and oxygen ?
 (a) F, N (b) N, Cl
 (c) N, P (d) N, C
23. The molecule exhibiting maximum number of non-bonding electron pairs (l.p.) around the central atom is :
 (a) $XeOF_4$ (b) XeO_2F_2
 (c) XeF_3^- (d) XeO_3
24. Which is the following pairs of species have identical shapes?
 (a) NO_2^+ and NO_2^- (b) PCl_5 and BrF_5
 (c) XeF_4 and ICl_4^- (d) $TeCl_4$ and XeO_4

Resonance

17. The correct order of increasing C–O bond strength of CO , CO_3^{2-} , CO_2 is :
 (a) $CO_3^{2-} < CO_2 < CO$ (b) $CO_2 < CO_3^{2-} < CO$
 (c) $CO < CO_3^{2-} < CO_2$ (d) $CO < CO_2 < CO_3^{2-}$
18. Resonance structures can be written for :
 (a) O_3 (b) NH_3
 (c) CH_4 (d) H_2O
25. Which is not correctly matched ?
 (a) XeO_3 – Trigonal bipyramidal
 (b) ClF_3 – bent T-shape
 (c) $XeOF_4$ – Square pyramidal
 (d) XeF_2 – Linear shape
26. The compound MX_4 is tetrahedral. The number of $\angle XMX$ angles in the compound is :
 (a) three (b) four
 (c) five (d) six

VSEPR

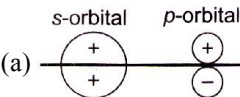
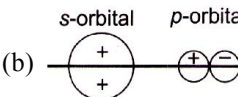
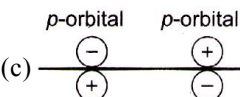
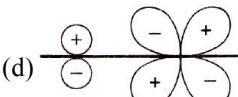
19. Correct statement regarding molecules SF_4 , CF_4 and XeF_4 are :
 (a) 2, 0 and 1 lone pairs of central atom respectively
 (b) 1, 0 and 1 lone pairs of central atom respectively
 (c) 0, 0 and 2 lone pairs of central atom respectively
 (d) 1, 0 and 2 lone pairs of central atom respectively
20. The pair of species with similar shape is :
 (a) PCl_3 , NH_3 (b) CF_4 , SF_4
 (c) $PbCl_2$, CO_2 (d) PF_5 , IF_5
21. The geometrical arrangement of orbitals and shape of I_3^- are respectively :
 (a) trigonal bipyramidal geometry, linear shape
 (b) hexagonal geometry, T-shape
 (c) triangular planar geometry, triangular shape
 (d) tetrahedral geometry, pyramidal shape
22. The shape of the noble gas compound XeF_4 is :
 (a) square planar (b) distorted tetrahedral
 (c) tetrahedral (d) octahedral
27. Which of the following are isoelectronic and isostructural ?
 NO_3^- , CO_3^{2-} , ClO_3^- , SO_3
 (a) NO_3^- , CO_3^{2-} (b) SO_3 , NO_3^-
 (c) ClO_3^- , CO_3^{2-} (d) CO_3^{2-} , ClO_3^-
28. In which of the following pairs, both the species have the same hybridisation ?
 (I) SF_4 , XeF_4 (II) I_3^- , XeF_2
 (III) ICl_4^+ , $SiCl_4$ (IV) ClO_3^- , PO_4^{3-}
 (a) I, II (b) II, III
 (c) II, IV (d) I, II, III
29. Which of the following molecule is theoretically not possible?
 (a) SF_4 (b) OF_2
 (c) OF_4 (d) O_2F_2

VBT

30. The strength of bonds formed by 2s-2s, 2p-2p and 2p-2s overlap has the order :
 (a) $s-s > p-p > p-s$ (b) $s-s > p-s > p-p$
 (c) $p-p > p-s > s-s$ (d) $p-p > s-s > p-s$
31. C_2H_2 is isostructural with :
 (a) H_2O_2 (b) NO_2
 (c) $SnCl_2$ (d) CO_2
32. When iodine is dissolved in aqueous potassium iodide, the shape of the species formed is :
 (a) linear (b) angular
 (c) triangular (d) see-saw
39. Select pair of compounds in which both have different hybridization but have same molecular geometry :
 (a) BF_3 , BrF_3 (b) ICl_2^0 , $BeCl_2$
 (c) BCl_3 , PCl_3 (d) PCl_3 , NCl_3
40. Which of the following statements is incorrect for sigma and pi-bonds formed between two carbon atoms ?
 (a) Sigma-bond is stronger than a pi-bond
 (b) Bond energies of sigma and pi-bonds are of the order of 264 kJ/mol and 347 kJ/mol
 (c) Free rotation of surrounding atoms about a sigma-bond is allowed but not in case of a pi-bond
 (d) Sigma-bond determines the direction between carbon atoms but a pi-bond has no primary effect in this regard

Hybridization

33. Which one of the following is the correct set with respect to molecule, hybridization and shape ?
 (a) $BeCl_2$, sp^2 , linear
 (b) $BeCl_2$, sp^2 , triangular planar
 (c) BCl_3 , sp^2 , triangular planar
 (d) BCl_3 , sp^3 , tetrahedral
34. The hybridization of the central atom in ICl_2^+ is :
 (a) dsp^2 (b) sp
 (c) sp^2 (d) sp^3
35. The state of hybridization of the central atom is not the same as in the others :
 (a) B in BF_3 (b) O in H_3O^+
 (c) N in NH_3 (d) P in PCl_3
36. Which of the following statements is incorrect for PCl_5 ?
 (a) Its three P-Cl bond lengths are equal
 (b) It involves sp^3d hybridization
 (c) It has linear geometry
 (d) Its shape is trigonal bipyramidal
37. The number of sp^2-s sigma bonds in benzene are :
 (a) 3 (b) 6
 (c) 12 (d) none of these
38. During the complete combustion of methane CH_4 , what change in hybridisation does the carbon atom undergo ?
 (a) sp^3 to sp (b) sp^3 to sp^2
 (c) sp^2 to sp (d) sp^2 to sp^3
41. Assuming the bond direction to the z-axis, which of the overlapping of atomic orbitals of two atom (A) and (B) will result in bonding ?
 (I) s-orbital of A and p_x orbital of B
 (II) s-orbital of A and p_z orbital of B
 (III) p_y -orbital of A and p_z orbital of B
 (IV) s-orbital of both (A) and (B)
 (a) I and IV (b) I and II
 (c) III and IV (d) II and IV
42. The hybridisation of atomic orbitals of central atom "Xe" in XeO_4 , XeO_2F_2 and $XeOF_4$ respectively.
 (a) sp^3 , sp^3d^2 , sp^3d^2 (b) sp^3d , sp^3d , sp^3d^2
 (c) sp^3 , sp^3d^2 , sp^3d (d) sp^3 , sp^3d , sp^3d^2
43. For exhibiting tetravalency, carbon atoms have to be excited. Now, which of the following statements is true ?
 (a) Excitation occurs before bonding
 (b) Bonding occurs before excitation
 (c) Both bonding and excitation occur simultaneously
 (d) Two bonds are formed first, then excitation occurs followed by formation of another two bonds
44. Which of the following statements is true about bonding and excitation ?
 (a) Energy required for excitation of carbon atoms (96 Kcal/mol) is less than energy released in bonding
 (b) Energy required for excitation is more than energy released in bonding
 (c) Energy required for excitation is equal to the energy released in bonding
 (d) None of these

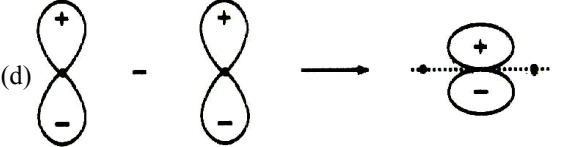
45. Which of the following statements is true about hybridization ?
- Only those atomic orbitals can be hybridized which do not differ much in shape.
 - Only those atomic orbitals can be hybridized which do not differ much in energy.
 - Only those atomic orbitals can be hybridized which do not differ much in size.
 - Only those atomic orbitals can be hybridized which do not differ much in overlap integrals.
46. Which of the following statements is true about hybridization ?
- Hybridization generates new set of atomic orbitals identical in shape but not in size and energy.
 - Hybridization generates new set of atomic orbitals identical in size but not in shape and energy.
 - Hybridization generates new set of atomic orbitals identical energy but not in shape and size.
 - Hybridization generates new set of atomic orbitals identical in shape, size and energy.
47. Which of the following statements is true about hybridization ?
- Hybrid orbitals frequently undergo linear overlaps making sigma bonds.
 - Hybrid orbitals frequently undergo lateral overlaps making π -bonds. In other words, there are several compounds in which π -bonds are formed using hybrid orbitals.
 - Hybrid orbitals are molecular orbitals.
 - A hybrid orbital bigger in size makes shorter bond.
48. In $2sp$ hybridization, $2s$ -orbital can be mixed with
- Only $2p_x$
 - Only $2p_y$
 - Only $2p_z$
 - Any one of $2p_x$, $2p_y$ and $2p_z$
49. In $2sp^2$ orbital, character of $2p_z$ orbital will be
- always 33.33%
 - always 0%
 - always 66.66%
 - either 33.33% or 0%
50. Shapes of sp^3 , sp^2 and sp orbitals are like
- 
 - 
 - 
 - 
51. Energy content of a molecule will be less if
- bond energy is more
 - bond energy is less
 - magnitude of overlap is very less
 - none of these
52. Which of the following orders is correct for electronegativity ?
- $sp^3C > sp^2C > spC$
 - $spC > sp^2C > sp^3C$
 - $sp^2C > spC > sp^3C$
 - $sp^3C > spC > sp^2C$
53. $CH_2 = C = CH_2$
In this molecule (allene)
- all three C-atoms are sp^2 hybridized
 - both terminal C-atoms are sp^2 hybridized while central C-atom is sp -hybridized
 - both terminal C-atoms are sp -hybridized while central C-atom is sp^2 hybridized
 - none of these
- MOT**
54. The nodal plane in the π -bond of ethene is located in :
- the molecular plane
 - a plane parallel to the molecule plane
 - a plane perpendicular to the molecular plane which bisects the carbon-carbon σ bond at right angle
 - a plane perpendicular to the molecular plane which contains the carbon-carbon bond
55. Which of the following leads to bonding ?
- 
 - 
 - 
 - 
56. Which of the following is paramagnetic ?
- O_2^{2-}
 - NO
 - CO
 - CN^-
57. The ground state electronic configuration of valence shell electrons in nitrogen molecule (N_2) is written ($\sigma 2s$)² ($\sigma^* 2s$)² ($\pi 2p$)⁴ ($\sigma 2p$)². Hence, the bond order in nitrogen molecule is :
- 0
 - 1
 - 2
 - 3

58. The ion that is isoelectronic with CO is :
(a) O^{2-} (b) N_2^+
(c) O^+ (d) CN^-
59. Which one of the following is not paramagnetic ?
(a) NO (b) N_2^+
(c) CO (d) O_2
60. When N_2 is formed from N_2^+ bond-order and when O_2 is formed from O_2^+ bond-order :
(a) increases (b) decreases
(c) increases, decreases (d) decreases, increases
61. Which of the following have identical bond order ?
(I) CN^- (II) O_2
(III) NO^+ (IV) O_2^-
(a) I, III (b) II, IV
(c) I, II, III (d) I, IV
62. Among unpaired electron is present in :
(a) KO_2 only (b) NO_2^+ and BaO_2
(c) KO_2 and AlO_2^- (d) BaO_2 only
63. Bond order of O_2 , O_2^+ , O_2^- and O_2^{2-} is in order :
(a) $O_2^- < O_2^{2-} < O_2 < O_2^+$ (b) $O_2^{2-} < O_2^- < O_2 < O_2^+$
(c) $O_2^+ < O_2 < O_2^- < O_2^{2-}$ (d) $O_2 < O_2^+ < O_2^- < O_2^{2-}$

COMPREHENSION (Ques. 64 to 68)

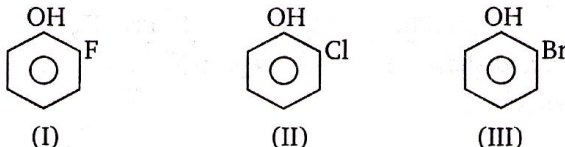
According to MOT, two atomic orbitals overlap resulting in the formation of molecular orbitals. Number of atomic orbitals overlapping together is equal to the molecular orbital formed. The two atomic orbital thus formed by LCAO (linear combination of atomic orbital) in the same phase or in the different phase are known as bonding and antibonding molecular orbitals respectively. The energy of bonding molecular orbital is lower than that of the pure atomic orbital by an amount Δ . This known as the stabilization energy. The energy of antibonding molecular orbital is increased by Δ' (destabilisation energy).

64. Which among the following pairs contain both paramagnetic species
(a) O_2^{2-} and N_2^- (b) O_2^- and N_2
(c) O_2 and N_2 (d) O_2 and N_2^-

65. Which of the following statement(s) is true
(a) Higher the bond order lesser the bond length
(b) Higher the bond order greater the bond length
(c) Higher the bond order lesser the bond energy
(d) Higher the bond order lesser the number of bonds
66. Which of the following pairs of molecule can exist ?
(a) He_2 and Be_2 (b) O_2^{2-} and Na_2
(c) O_2^{2-} and H_2^{2-} (d) Be_2 and Mg_2
67. Which of the following combination of orbitals is correct ?
(a) 
(b) 
(c) 
(d) 
68. Which of the following statements is not correct regarding bonding molecular orbitals ?
(a) Bonding molecular orbitals possess less energy than the atomic orbitals from which they are formed
(b) Bonding molecular orbitals have low electron density between the two nuclei
(c) Electron in bonding molecular contributes to the attraction between atoms
(d) They are formed when the lobes of the combining atomic orbitals have the same sign

Properties of Covalent cmpds

69. The BCl_3 is a planar molecule whereas NCl_3 is pyramidal because :
(a) B-Cl bond is more polar than N-Cl bond
(b) N-Cl bond is more covalent than B-Cl bond
(c) nitrogen atom is smaller than boron atom
(d) BCl_3 has no lone pair electrons but NCl_3 has a lone pair of electrons
70. The correct order of the O-O bond length in O_2 , H_2O_2 and O_3 is :
(a) $O_2 > O_3 > H_2O_2$ (b) $O_3 > H_2O_2 > O_2$
(c) $O_2 > H_2O_2 > O_3$ (d) $H_2O_2 > O_3 > O_2$

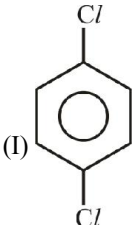
71. The structure and hybridisation of $\text{Si}(\text{CH}_3)_4$ is :
(a) bent, sp (b) trigonal, sp^2
(c) octahedral, sp^3d (d) tetrahedral, sp^3
72. In which of the following bond angle is maximum :
(a) NH_3 (b) NH_4^+
(c) PCl_3 (d) SCl_2
73. BrF_3 molecule, the lone pairs occupy equatorial position to minimize.
(a) lone pair-bond pair repulsion only
(b) bond pair-bond pair repulsion only
(c) lone pair-lone pair repulsion and lone pair-bond pair repulsion
(d) lone pair-lone pair repulsion only
74. H_2O is dipolar, whereas BeF_2 is not. It is because :
(a) the electronegativity of F is greater than that of O
(b) H_2O involves hydrogen bonding whereas BeF_2 is a discrete molecule
(c) H_2O is linear and BeF_2 is angular
(d) H_2O is angular and BeF_2 is linear
75. The ratio of π and σ bonds in benzene is :
(a) 1 : 2 (b) 1 : 4
(c) 1 : 5 (d) 1 : 6
76. In graphite each carbon atom is :
(a) sp^2 hybridised (b) sp^3 hybridised
(c) sp hybridised (d) not hybridised
77. Both BF_3 and NF_3 are covalent but BF_3 molecule is non-polar while NF_3 is polar because :
(a) atomic size of boron is smaller than nitrogen
(b) BF_3 is planar but NF_3 is pyramidal
(c) boron is a metal while nitrogen is gas
(d) BF bond has no dipole moment while NF bond has dipole
78. The highest dipole moment is of :
(a) CF_4 (b) CH_3OH
(c) CO_2 (d) CH_3F
79. Which one of the following is the most polar bond ?
(a) N–F (b) N–N
(c) N–Cl (d) O–F
80. The hybridisation of atomic orbitals of N in NO_2^+ , NO_3^- and NH_4^+ are respectively :
(a) sp, sp^2 , sp^3 (b) sp, sp^3 , sp^2
(c) sp^2 , sp, sp^3 (d) sp^2 , sp^3 , sp
81. Which of the following molecules does not have a dipole moment ?
(a) IBr (b) CHCl_3
(c) CH_2Cl_2 (d) BF_3
- ### Hydrogen Bonding
82. The high density of water compared to ice is due to :
(a) hydrogen bonding interactions
(b) dipole-dipole interactions
(c) dipole-induced dipole interactions
(d) induced dipole induced dipole interactions
83. Which one of the following molecules will form a linear polymeric structure due to hydrogen bonding ?
(a) NH_3 (b) H_2O
(c) HCl (d) HF
84. The boiling point of alcohol is higher than ether due to :
(a) hydrogen bonding
(b) large size of alcohol
(c) presence of –OH group
(d) high molecular weight
85. The correct order of boiling point is :

(a) I > II > III (b) III > II > I
(c) II > I > III (d) III > I > II
86. The boiling points at atmospheric pressure of HF, H_2S , NH_3 can be arranged in the following order :
(a) $\text{HF} > \text{NH}_3 > \text{H}_2\text{S}$ (b) $\text{HF} > \text{H}_2\text{S} > \text{NH}_3$
(c) $\text{HF} < \text{H}_2\text{S} < \text{NH}_3$ (d) $\text{HF} < \text{NH}_3 < \text{H}_2\text{S}$
87. Which statement is correct ?
(a) m.p. of H_2O , NH_3 , HF are maximum in their respective group due to intermolecular H-bonding
(b) b.p. of CH_4 out of CH_4 , SiH_4 , GeH_4 and SnH_4 is least due to weak intermolecular force of attraction
(c) formic acid forms dimer by H-bonding
(d) all are correct

88. Which of the following is not true about H_2O molecule ?
- The molecule has $\mu = 0$
 - The molecule can act as a base
 - Shows abnormally high boiling point in comparison to the hydrides of other elements of oxygen group
 - The molecule has a bent shape
89. Which of the following does not contain any co-ordinate bond ?
- H_3O^+
 - BF_4^-
 - HF_2^-
 - NH_4^+
90. The boiling points of methanol, water and dimethyl ether are respectively 65°C , 100°C and 34.5°C . Which of the following best explains these wide variations in b.p. ?
- The molecular mass increases from water (18) to methanol (32) to diethyl ether (74)
 - The extent of H-bonding decreases from water to methanol while it is absent in ether
 - The extent of intramolecular H-bonding decreases from ether to methanol to water
 - The density of water is 1.00 g mL^{-1} , methanol 0.7914 g mL^{-1} and that of diethyl ether is 0.7137 g mL^{-1}
91. In ice, the length of H-bonds :
- is less than that of covalent bonds
 - is greater than that of covalent bonds
 - is same as that of covalent bonds
 - can be less greater or same as that of covalent bonds
92. The correct order of the strength of H-bonds is :
- $\text{H}\cdots\text{F} > \text{H}\cdots\text{O} > \text{H}\cdots\text{N}$
 - $\text{H}\cdots\text{N} > \text{H}\cdots\text{O} > \text{H}\cdots\text{F}$
 - $\text{H}\cdots\text{O} > \text{H}\cdots\text{N} > \text{H}\cdots\text{F}$
 - $\text{H}\cdots\text{F} > \text{H}\cdots\text{N} > \text{H}\cdots\text{O}$
93. o-nitrophenol can be easily steam distilled whereas p-nitrophenol cannot be. This is because of :
- strong intermolecular hydrogen bonding in o-nitrophenol
 - strong intramolecular hydrogen bonding in o-nitrophenol
 - strong intramolecular hydrogen bonding in p-nitrophenol
 - dipole moment of p-nitrophenol is larger than that of o-nitrophenol
94. What is not true about ice ?
- It has open cage like structure
 - It has less density than water
 - Each O atom is surrounded by 4 H atoms
 - Each O atom has four H-bonds around it

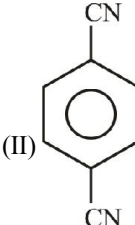
EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTION

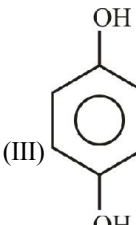
1. Hybridisation of the underline atom changes in : (2002)
(a) $\underline{\text{Al}}\text{H}_3$ changes of AlH_4^- (b) $\text{H}_2\underline{\text{O}}$ changes to H_3O^+
(c) $\underline{\text{N}}\text{H}_3$ changes to NH_4^+ (d) in all cases
2. In which of the following species the interatomic bond angle is $109^\circ 28'$? (2002)
(a) NH_3 , $(\text{BF}_4)^{-1}$ (b) $(\text{NH}_4)^+$, BF_4^-
(c) NH_3 , BF_4 (d) $(\text{NH}_2)^{-1}$, BF_3 .
3. Increasing order of bond strength of O_2 , O_2^- , O_2^{2-} and O_2^+ is (2002)
(a) $\text{O}_2^+ < \text{O}_2 < \text{O}_2^- < \text{O}_2^{2-}$ (b) $\text{O}_2 < \text{O}_2^+ < \text{O}_2^- < \text{O}_2^{2-}$
(c) $\text{O}_2^- < \text{O}_2^{2-} < \text{O}_2^+ < \text{O}_2$ (d) $\text{O}_2^{2-} < \text{O}_2^- < \text{O}_2 < \text{O}_2^+$
4. Number of P–O bonds in P_4O_{10} is (2002)
(a) 17 (b) 16
(c) 15 (d) 6
5. Bond angle of $109^\circ 28'$ is found in
(a) NH_3 (b) H_2O
(c) CH_5^+ (d) NH_4^+
6. The number of lone pairs on Xe in XeF_2 , XeF_4 and XeF_6 respectively, are (2002)
(a) 3, 2, 1 (b) 2, 4, 6
(c) 1, 2, 3 (d) 6, 4, 2
7. Select correct statement. (2002)
(a) When a covalent bond is formed, transfer of electrons takes place
(b) Pure H_2O does not contain any ion
(c) A bond is formed when attractive forces overcome repulsive forces
(d) HF is less polar than HBr
8. The pair of species having identical shapes for molecules of both species is (2003)
(a) XeF_2 , CO_2 (b) BF_3 , PCl_3
(c) PF_5 , IF_5 (d) CF_4 , SF_4
9. Which one of the following compounds has the smallest bond angle ? (2003)
(a) H_2O (b) H_2S
(c) NH_3 (d) SO_2
10. An ether is more volatile than an alcohol having the same molecular formula. This is due to (2003)
(a) alcohols having resonance structures
(b) inter-molecular hydrogen bonding in ethers
(c) inter-molecular hydrogen bonding in alcohols
(d) dipolar character of ethers
11. Which one of the following pairs of molecules will have permanent dipole moments for both members ? (2003)
(a) SiF_4 and NO_2 (b) NO_2 and CO_2
(c) NO_2 and O_3 (d) SiF_4 and CO_2
12. The pair of species having identical shapes for molecules of both species, is (2003)
(a) CF_4 , SF_4 (b) XeF_2 , CO_2
(c) BF_3 , PCl_3 (d) PF_5 , IF_5
13. The bond order in NO is 2.5 while that in NO^+ is 3. Which of the following statements is true for these two species ? (2004)
(a) Bond length in NO^+ is greater than in NO
(b) Bond length in NO is greater than in NO^+
(c) Bond length in NO^+ is equal to that in NO
(d) Bond length is unpredictable
14. The states of hybridisation of boron and oxygen atoms in boric acid (H_3BO_3) are respectively (2004)
(a) sp^2 and sp^2 (b) sp^2 and sp^3
(c) sp^3 and sp^2 (d) sp^3 and sp^3
15. The correct order of bond angles (smallest first) in H_2S , NH_3 , BF_3 and SiH_4 is (2004)
(a) $\text{H}_2\text{S} < \text{SiH}_4 < \text{NH}_3 < \text{BF}_3$
(b) $\text{NH}_3 < \text{H}_2\text{S} < \text{SiH}_4 < \text{BF}_3$
(c) $\text{H}_2\text{S} < \text{NH}_3 < \text{SiH}_4 < \text{BF}_3$
(d) $\text{H}_2\text{S} < \text{NH}_3 < \text{BF}_3 < \text{SiH}_4$

16. Which one of the following has the regular tetrahedral structure ? (2004)
(Atomic number : B = 5, S = 16, Ni = 28, Xe = 54)
(a) XeF_4 (b) SF_4
(c) BF_4^- (d) $[\text{Ni}(\text{CN})_4]^{2-}$
17. The maximum number of 90° angles between bond pair-bond pair of electrons is observed in (2004)
(a) dsp^3 hybridisation (b) sp^3d hybridisation
(c) dsp^2 hybridisation (d) sp^3d^2 hybridisation
18. Lattice energy of an ionic compound depends upon (2005)
(a) Charge on the ion and size of the ion
(b) Packing of ions only
(c) Size of the ion only
(d) Charge on the ion only
19. The molecular shapes of SF_4 , CF_4 and XeF_4 are (2005)
(a) different with 1, 0 and 2 lone pair of electrons on the central atoms, respectively
(b) different with 0, 1 and 2 lone pair of electrons on the central atoms, respectively.
(c) the same with 1, 1 and 1 lone pair of electrons on the central atoms, respectively
(d) the same with 2, 0 and 1 lone pair of electrons on the central atoms, respectively
20. Which one of the following species is diamagnetic in nature ? (2005)
(a) H_2^- (b) H_2^+
(c) H_2 (d) He_2^+
21. The decreasing values of bond angles from NH_3 (106°) to SbH_3 (101°) down group-15 of the periodic table is due to (2006)
(a) decreasing lp-bp repulsion
(b) decreasing electronegativity
(c) increasing bp-bp repulsion
(d) increasing p-orbital character in sp^3
22. In which of the following molecules/ions are all the bonds not equal ? (2006)
(a) XeF_4 (b) BF_4^-
(c) SF_4 (d) SiF_4
23. Which of the following molecules/ions does not contain unpaired electrons ? (2006)
(a) N_2^+ (b) O_2
(c) O_2^{2-} (d) B_2
24. Which one of the following pairs of species have the same bond order ? (2007)
(a) CN^- and NO^+ (b) CN^- and CN^+
(c) O_2^- and CN^- (d) NO^+ and CN^+
25. In which of the following ionization processes, the bond order has increased and the magnetic behaviour has changed ? (2007)
(a) $\text{N}_2 \rightarrow \text{N}_2^+$ (b) $\text{C}_2 \rightarrow \text{C}_2^+$
(c) $\text{NO} \rightarrow \text{NO}^+$ (d) $\text{O}_2 \rightarrow \text{O}_2^+$
26. Which of the following species exhibits diamagnetic behaviour ? (2007)
(a) O_2^{2-} (b) O_2^+
(c) O_2 (d) NO
27. Which of the following hydrogen bonds is the strongest ? (2007)
(a) $\text{O}-\text{H}\cdots\text{N}$ (b) $\text{F}-\text{H}\cdots\text{F}$
(c) $\text{O}-\text{H}\cdots\text{O}$ (d) $\text{O}-\text{H}\cdots\text{F}$
28. Which one of the following constitutes a group of the isoelectronic species ? (2008)
(a) C_2^{2-} , O_2^- , CO , NO (b) NO^+ , C_2^{2-} , CN^- , N_2
(c) CN^- , N_2 , O_2^{2-} , CO_2^{2-} (d) N_2 , O_2^- , NO^+ , CO
29. Using MO theory, predict which of the following species has the shortest bond length ? (2008)
(a) O_2^+ (b) O_2^-
(c) O_2^{2-} (d) O_2^{2+}
30. The bond dissociation energy of B – F in BF_3 is 646 kJ mol^{-1} . The reason for higher B – F bond dissociation energy as compared to that of C – F is (2008)
(a) stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4 .
(b) significant $\text{p}\pi-\text{p}\pi$ interaction between B and F in BF_3 whereas there is no possibility of such interaction between C and F in CF_4 .
(c) lower degree of $\text{p}\pi-\text{p}\pi$ interaction between B and F in BF_3 than that between C and F in CF_4 .
(d) smaller size of B-atom as compared to that of C-atom.

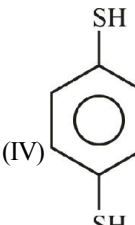
31. The number of types of bonds between two carbon atoms in calcium carbide is (2011)
(a) one sigma, two pi (b) one sigma, one pi
(c) two sigma, one pi (d) two sigma, two pi
32. Which of the following has maximum number of lone pairs associated with Xe ? (2011)
(a) XeO_3 (b) XeF_4
(c) XeF_6 (d) XeF_2
33. The structure of IF_7 is (2011)
(a) square pyramid (b) trigonal bipyramid
(c) octahedral (d) pentagonal bipyramid
34. The hybridization of orbitals of N atom in NO_3^- , NO_2^+ and NH_4^+ are respectively (2011)
(a) sp , sp^2 , sp^3 (b) sp^2 , sp , sp^3
(c) sp , sp^3 , sp^2 (d) sp^2 , sp^3 , sp
35. Among the following the maximum covalent character is shown by the compound (2011)
(a) FeCl_2 (b) SnCl_2
(c) AlCl_3 (d) MgCl_2
36. In which of the following pairs the two species are not isostructural ? (2012)
(a) CO_3^{2-} and NO_3^- (b) PCl_4^+ and SiCl_4
(c) PF_5 and BrF_5 (d) AlF_6^{3-} and SF_6
37. The molecule having smallest bond angle is (2012)
(a) NCl_3 (b) AsCl_3
(c) SbCl_3 (d) PCl_3
38. Stability of the species Li_2 , Li_2^- and Li_2^+ increases in the order of (2012)
(a) $\text{Li}_2 < \text{Li}_2^+ < \text{Li}_2^-$ (b) $\text{Li}_2^- < \text{Li}_2^+ < \text{Li}_2$
(c) $\text{Li}_2 < \text{Li}_2^- < \text{Li}_2^+$ (d) $\text{Li}_2^- < \text{Li}_2 < \text{Li}_2^+$
39. In which of the following pairs of molecules/ions both the species are not likely to exist ? (2013)
(a) H_2^+ , He_2^{2-} (b) H_2^- , He_2^{2-}
(c) H_2^{2+} , He_2 (d) H_2^- , He_2^{2+}
40. Which one of the following molecules is expected to exhibit diamagnetic behaviour ? (2013)
(a) C_2 (b) O_2
(c) S_2 (d) None of these
41. For which of the following molecule significant $\mu \neq 0$? (2014)
- 

(I)



(II)
- 

(III)



(IV)
- (a) (I) and (II) (b) Only (III)
(c) (III) and (IV) (d) Only (I)
42. The species in which the N atom is in a state of sp hybridization is: (2016)
(a) NO_2^- (b) NO_3^-
(c) NO_2 (d) NO_2^+
43. Which of the following species is not paramagnetic ? (2017)
(a) NO (b) CO
(c) O_2 (d) B_2
44. Total number of lone pair of electrons in I_3^- ions is: (2018)
(a) 12 (b) 3
(c) 6 (d) 9
45. Which of the following compounds contain(s) no covalent bond(s)? (2018)
 KCl , PH_3 , O_2 , B_2H_6 , H_2SO_4
(a) KCl , B_2H_6 (b) KCl , B_2H_6 , PH_3
(c) KCl , H_2SO_4 (d) KCl
46. According to molecular orbital theory, which of the following will not be a viable molecule? (2018)
(a) H_2^{2-} (b) He_2^{2+}
(c) He_2^+ (d) H_2^-

JEE MAINS ONLINE QUESTION

1. Which of the following has unpaired electron(s)?

Online 2014 SET (1)

- (a) N_2 (b) O_2^-
(c) N_2^{2+} (d) O_2^{2+}

2. Which one of the following does not have a pyramidal shape?

Online 2014 SET (2)

- (a) $(CH_3)_3N$ (b) $P(CH_3)_3$
(c) $P(SiH_3)_3$ (d) $(SiH_3)_3N$

3. Shapes of certain interhalogen compounds are stated below. Which one of them is not correctly stated?

Online 2014 SET (2)

- (a) IF_7 : pentagonal bipyramid
(b) ICl_3 : planar dimeric
(c) BrF_3 : planar T-shaped
(d) BrF_5 : trigonal bipyramid

4. In allene (C_3H_4), the type(s) of hybridization of the carbon atoms is (are):

Online 2014 SET (2)

- (a) sp^2 and sp^3 (b) sp and sp^3
(c) sp^2 and sp (d) only sp^2

5. The correct order of bond dissociation energy among N_2 , O_2^- , O_2^- is shown in which of the following arrangements?

Online 2014 SET (2)

- (a) $N_2 > O_2 > O_2^-$ (b) $O_2 > O_2^- > N_2$
(c) $N_2 > O_2^- > O_2$ (d) $O_2^- > O_2 > N_2$

6. Which of the following molecules has two sigma (σ) and two pi (π) bonds?

Online 2014 SET (3)

- (a) $C_2H_2Cl_2$ (b) HCN
(c) N_2F_2 (d) C_2H_4

7. Amongst $LiCl$, $RbCl$, $BeCl_2$ and $MgCl_2$ the compounds with the greatest and the least ionic character, respectively are :

Online 2014 SET (4)

- (a) $LiCl$ and $RbCl$ (b) $MgCl_2$ and $BeCl_2$
(c) $RbCl$ and $BeCl_2$ (d) $RbCl$ and $MgCl_2$

8. Which of the alkaline earth metal halides given below is essentially covalent in nature?

Online 2015 SET (1)

- (a) $SrCl_2$ (b) $CaCl_2$
(c) $BeCl_2$ (d) $MgCl_2$

9. The group of molecules having identical shape is :

Online 2016 SET (1)

- (a) SF_4 , XeF_4 , CCl_4
(b) ClF_3 , $XeOF_2$, XeF_3^+
(c) BF_3 , PCl_3 , XeO_3
(d) PCl_5 , IF_5 , XeO_2F_2

10. The bond angle H-X-H is the greatest in the compound :

Online 2016 SET (2)

- (a) CH_4 (b) NH_3
(c) H_2O (d) PH_3

11. Which of the following is paramagnetic?

Online 2017 SET (1)

- (a) NO^+ (b) CO
(c) O_2^{2-} (d) B_2

12. sp^3d^2 hybridization is not displayed by :

Online 2017 SET (1)

- (a) BrF_5 (b) SF_6
(c) $[CrF_6]^{3-}$ (d) PF_5

13. The group having triangular planar structures is :

Online 2017 SET (2)

- (a) BF_3 , NF_3 , CO_3^{2-}
(b) CO_3^{2-} , NO_3^- , SO_3
(c) NH_3 , SO_3 , CO_3^{2-}
(d) NCl_3 , BCl_3 , SO_3

14. In the molecular orbital diagram for the molecular ion, N_2^+ , the number of electrons in the σ_{2p} molecular orbital is :

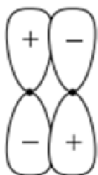
Online 2018 SET (1)

- (a) 0 (b) 1
(c) 2 (d) 3

15. Identify the pair in which the geometry of the species is T-shape and squarepyramidal, respectively :

Online 2018 SET (1)

- (a) ClF_3 and IO_4^-
(b) ICl_2^- and ICl_5
(c) $XeOF_2$ and $XeOF_4$
(d) IO_3^- and $IO_2F_2^-$

16. The decreasing order of bond angles in BF_3 , NH_3 , PF_3 and I_3 is :
(a) $\text{I}_3 > \text{NH}_3 > \text{PF}_3 > \text{BF}_3$ (b) $\text{I}_3 > \text{BF}_3 > \text{NH}_3 > \text{PF}_3$
(c) $\text{BF}_3 > \text{I}_3 > \text{PF}_3 > \text{NH}_3$ (d) $\text{BF}_3 > \text{NH}_3 > \text{PF}_3 > \text{I}_3$
(Online 2018 SET (1))
17. Which of the following best describes the diagram below of a molecular orbital ?

(a) A non-bonding orbital
(b) An antibonding σ orbital
(c) A bonding π orbital
(d) An antibonding orbital
(Online 2018 SET (2))
18. In XeO_3F_2 , the number of bond pair(s), π -bond(s) and lone pair(s) on Xe atom respectively are :
(a) 5, 2, 0 (b) 4, 2, 2
(c) 5, 3, 0 (d) 4, 4, 0
(Online 2018 SET (2))
19. The incorrect geometry is represented by :
(a) BF_3 - trigonal planar
(b) H_2O - bent
(c) NF_3 - trigonal planar
(d) AsF_5 - trigonal bipyramidal
(Online 2018 SET (3))
20. Which of the following conversions involves change in both shape and hybridisation ?
(a) $\text{NH}_3 \rightarrow \text{NH}_4^+$ (b) $\text{CH}_4 \rightarrow \text{C}_2\text{H}_6$
(c) $\text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+$ (d) $\text{BF}_3 \rightarrow \text{BF}_4$
(Online 2018 SET (3))

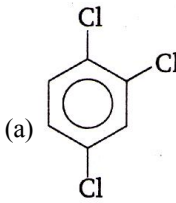
EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

1. All questions marked “S” are single choice questions
2. All questions marked “M” are multiple choice questions
3. All questions marked “C” are comprehension based questions
4. All questions marked “A” are assertion–reason type questions
 - (A) If both assertion and reason are correct and reason is the correct explanation of assertion.
 - (B) If both assertion and reason are true but reason is not the correct explanation of assertion.
 - (C) If assertion is true but reason is false.
 - (D) If reason is true but assertion is false.
5. All questions marked “X” are matrix–match type questions
6. All questions marked “I” are integer type questions

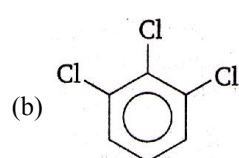
Ionic Bonds

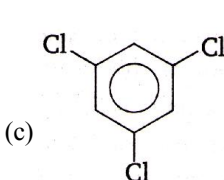
1. (S) The ionic bonds X^+Y^- are formed when :
 - (I) electron affinity of Y is high
 - (II) ionization energy of X is low
 - (III) lattice energy of XY is high
 - (IV) lattice energy of XY is lowChoose the correct code :
 - (a) I & II
 - (b) I & III
 - (c) I, II & III
 - (d) All
2. (S) Bonds present in N_2O_5 (nitrogen pentaoxide) are :
 - (a) only ionic
 - (b) only covalent
 - (c) covalent and co-ordinate
 - (d) covalent and ionic
3. (S) Among the oxides of nitrogen, N_2O , NO and NO_2 , molecules with unpaired electrons are :
 - (a) N_2O and NO
 - (b) NO and NO_2
 - (c) N_2O and NO_2
 - (d) NO_2 and its dimer
4. (S) The lowest O – O bond length in the following molecule is :
 - (a) O_2F_2
 - (b) O_2
 - (c) H_2O_2
 - (d) O_3
5. (S) What is the state of hybridisation of Xe in cationic part of solid XeF_6 ?
 - (a) sp^3d^3
 - (b) sp^3d^2
 - (c) sp^3d
 - (d) sp^3
6. (S) Choose the correct code of characteristics for the given order of hybrid orbitals of same atom,
 $sp < sp^2 < sp^3$
 - (i) Electronegativity
 - (ii) Bond angle between same hybrid orbitals
 - (iii) Size
 - (iv) Energy level
 - (a) (ii), (iii) and (iv)
 - (b) (iii), (iv)
 - (c) (ii) and (iv)
 - (d) (i), (ii), (iii) and (iv)
7. (S) Assuming pure 2s and 2p orbitals of carbon are used in forming CH_4 molecule, which of the following statement is false ?
 - (a) Three C – H bonds will be at right angle
 - (b) One C – H bond will be weaker than other three C – H bond
 - (c) The shape of molecule will be tetrahedral
 - (d) The angle of C – H bond formed by s - s overlapping will be uncertain with respect to other three bonds.
8. (S) Which is correct statements ?
As the s-character of a hybrid orbital decreases
 - (I) The bond angle decreases
 - (II) The bond strength increases
 - (III) The bond length increases
 - (IV) Size of orbitals increases
 - (a) (I), (III) and (IV)
 - (b) (II), (III) and (IV)
 - (c) (I) and (II)
 - (d) All are correct

9. (S) The state of hybridisation of central atom in dimer of BH_3 and BeH_2 is :
 (a) sp^2 , sp^2 (b) sp^3 , sp^2
 (c) sp^3 , sp^3 (d) sp^2 , sp^3
10. (S) The correct order of increasing s-character (in percentage) in the hybrid orbitals of following molecules/ions is :
 (I) CO_3^{2-} (II) XeF_4
 (III) I_3^- (IV) NCl_3
 (V) BeCl_2
 (a) $\text{II} < \text{III} < \text{IV} < \text{I} < \text{V}$ (b) $\text{II} < \text{IV} < \text{III} < \text{V} < \text{I}$
 (c) $\text{III} < \text{II} < \text{I} < \text{V} < \text{IV}$ (d) $\text{II} < \text{IV} < \text{III} < \text{I} < \text{V}$
11. (S) The shape of MnO_4^- ion and the hybridisation of Mn in MnO_4^- is :
 (a) tetrahedral, sp^3 (b) tetrahedral, d^3s
 (c) square planar, dsp^2 (d) square planar, sp^3
12. (S) The geometry of ammonia molecule can be best described as :
 (a) Nitrogen at one vertex of a regular tetrahedron, the other three vertices being occupied by three hydrogens
 (b) Nitrogen at the centre of the tetrahedron, three of the vertices being occupied by three hydrogens
 (c) Nitrogen at the centre of an equilateral triangle, three corners being occupied by three hydrogens
 (d) Nitrogen at the junction of a T, three open ends being occupied by three hydrogens
13. (S) Which molecular geometry is least likely to result from a trigonal bipyramidal electron geometry ?
 (a) Trigonal planar (b) See-saw
 (c) Linear (d) T-shaped
14. (S) Given the correct order of initials T or F for following statements. Use T if statement is true and F if it is false :
 (I) The order of repulsion between different pair of electrons is $l_p - l_p > l_p - b_p > b_p - b_p$
 (II) In general, as the number of lone pair of electrons on central atom increases, value of bond angle from normal bond angle also increases.
 (III) The number of lone pair on O in H_2O is 2 while on N in NH_3 is 1
 (IV) The structures of xenon fluorides and xenon oxyfluorides could not be explained on the basis of VSEPR theory
 (a) TTTF (b) TFTF
 (c) TFFT (d) TFFF
15. (S) Which one of the following compounds has the smallest bond angle ?
 (a) OH_2 (b) SH_2
 (c) NH_3 (d) SO_2
16. (S) Maximum bond angle is present in :
 (a) BBr_3 (b) BCl_3
 (c) BF_3 (d) none of these
17. (S) The correct order of H – M – H bonds angle is :
 (a) $\text{NH}_3 < \text{PH}_3 < \text{SbH}_3 < \text{BiH}_3$
 (b) $\text{AsH}_3 < \text{SbH}_3 < \text{PH}_3 < \text{NH}_3$
 (c) $\text{NH}_3 < \text{PH}_3 < \text{BiH}_3 < \text{SbH}_3$
 (d) $\text{BiH}_3 < \text{SbH}_3 < \text{AsH}_3 < \text{PH}_3$
18. (S) The correct increasing order of adjacent bond angle among BF_3 , PF_3 and ClF_3 :
 (a) $\text{BF}_3 < \text{PF}_3 < \text{ClF}_3$ (b) $\text{PF}_3 < \text{BF}_3 < \text{ClF}_3$
 (c) $\text{ClF}_3 < \text{PF}_3 < \text{BF}_3$ (d) $\text{BF}_3 = \text{PF}_3 = \text{ClF}_3$
19. (S) The bond angle of NH_3 , NH_4^+ and NH_2^- are in the order :
 (a) $\text{NH}_2^- > \text{NH}_3 > \text{NH}_4^+$
 (b) $\text{NH}_4^+ > \text{NH}_3 > \text{NH}_2^-$
 (c) $\text{NH}_3 > \text{NH}_2^- > \text{NH}_4^+$
 (d) $\text{NH}_3 > \text{NH}_4^+ > \text{NH}_2^-$
20. (S) The H – C – H bond angle in CH_4 is 109.5° , due to lone pair repulsion, the H – O – H angle in H_2O will :
 (a) remain the same (b) increase
 (c) decrease (d) become 180°
21. (S) The molecule having the largest bond angle is :
 (a) H_2O (b) H_2S
 (c) H_2Se (d) H_2Te
- Comprehension**
 The space model which is obtained by joining the points representing various bonded atoms gives the shape of the molecule. The geometry of the molecule is definite relative arrangement of the bonded atoms in a molecule. The shape and geometry of a molecule is explained by valence shell electron pair repulsion theory given by Gillespie and Nyholm.
22. (C) Select the correct code for the following repulsion orders, according to VSEPR theory :
 (I) lone pair-lone pair > lone pair-bond pair
 (II) lone pair-bond pair > bond pair-bond pair
 (III) lone pair-lone pair > bond pair-bond pair
 (IV) lone pair-bond pair > lone pair-bond pair
 (a) I, II & III (b) II & IV
 (c) I, II & IV (d) All

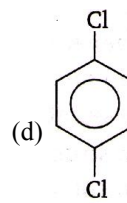
- 23. (C)** Which molecule has both shape and geometry identical ?
 (I) SnCl_2 (II) NH_3
 (III) PCl_5 (d) SF_6
 (a) I, III & IV (b) II, III & IV
 (c) III & IV (d) All
- 24. (C)** Which is not the electron geometry of covalent molecules ?
 (a) Pentagonal bipyramidal
 (b) Octahedral
 (c) Hexagonal (d) Tetrahedral
- 25. (S)** Which of the following compounds have the same no. of lone pairs with their central atom ?
 (I) XeF_5^- (II) BrF_3
 (III) XeF_2 (IV) H_3S^+
 (a) IV and V (b) I and III
 (c) I and II (d) II and IV
- 26. (S)** The correct order of bond angles is :
 (a) $\text{NO}_2^- > \text{NO}_2^+ > \text{NO}_2$
 (b) $\text{NO}_2^+ > \text{NO}_2^- > \text{NO}_2$
 (c) $\text{NO}_2 > \text{NO}_2^+ > \text{NO}_2^-$
 (d) $\text{NO}_2^+ > \text{NO}_2 > \text{NO}_2^-$
- 27. (S)** The geometry with respect to the central atom of the following molecules are :
 $\text{N}(\text{SiH}_3)_3$; Me_3N ; $(\text{SiH}_3)_3\text{P}$
 (a) planar, pyramidal, planar
 (b) planar, pyramidal, pyramidal
 (c) pyramidal, pyramidal, pyramidal
 (d) pyramidal, planar, pyramidal
- Comprehension**
 According to V.B.T., atoms of element form bond only to pair up their unpaired electrons present in ground state or excited state. This pairing of unpaired electron will take place by overlapping of orbitals each one having one unpaired electron with opposite spin.
- 28. (C)** Which of the following orbital combination does not form π -bond ?
 (a) $p_x + p_x$ sideways overlapping
 (b) $d_{x^2-y^2} + p_y$ sideways overlapping
 (c) $d_{xy} + d_{xy}$ sideways overlapping
 (d) $d_{yz} + p_y$ sideways overlapping
- 29. (C)** Which of the following orbital cannot form δ -bond ?
 (a) $d_{x^2-y^2}$ orbital (b) d_{xy} orbital
 (c) d_{z^2} orbital (d) d_{zx} orbital
- 30. (C)** Which of the following combination of orbitals does not form any type covalent bond (if z-axis is molecular axis) ?
 (a) $p_z + p_z$ (b) $p_y + p_y$
 (c) $s + p_y$ (d) $s + s$
- 31. (S)** Dipole moment of NF_3 is smaller than :
 (a) NH_3 (b) CO_2
 (c) BF_3 (d) CCl_4
- 32. (S)** Which of the following molecules will have polar bonds but zero dipole moment ?
 (a) O_2 (b) CHCl_3
 (c) CF_4 (d) none of these
- 33. (S)** Which has maximum dipole moment ?
- 

(a)

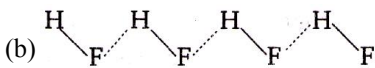
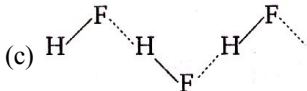
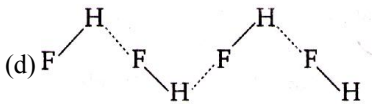


(b)
- 

(c)



(d)
- 34. (S)** The dipole moment of o, p and m-dichlorobenzene will be in the order :
 (a) $o > p > m$ (b) $p > o > m$
 (c) $m > o > p$ (d) $o > m > p$
- 35. (S)** Which of the following molecules has highest dipole moment?
 (a) BF_3 (b) NH_3
 (c) NF_3 (d) B_2H_6
- 36. (S)** Among the following, the molecule with highest dipole moment is :
 (a) CH_3Cl (b) CH_2Cl_2
 (c) CHCl_3 (d) CCl_4
- 37. (S)** Which of the following compounds has dipole moment approximately equal to that of chlorobenzene ?
 (a) o-dichlorobenzene (b) m-dichlorobenzene
 (c) p-dichlorobenzene (d) p-chloronitrobenzene

- 38. (S)** The correct sequence of polarity of the following molecule
- (1) Benzene (2) Inorganic Benene
(3) PCl_3F_2 (4) PCl_2F_3
- 1 2 3 4 1 2 3 4
- (a) P NP NP P (b) NP NP NP P
(c) NP P NP P (d) NP P P NP
(Where, P = polar, NP = non-polar)
- 39. (S)** The H-bonds in solid HF can be best represented as :
- (a) $\text{H} - \text{F} \cdots \text{H} - \text{F} \cdots \text{H} - \text{F}$
- (b) 
- (c) 
- (d) 
- 40. (S)** Which molecule geometry is least likely to result from a trigonal bipyramidal electron geometry ?
- (a) Trigonal planar (b) See-saw
(c) Linear (d) T-shaped
- 41. (M)** Select correct statement(s) regarding σ and π -bonds :
- (a) σ -bond lies on the line joining the nuclei of bonded atoms
(b) π -electron cloud lies on either side to the line joining the nuclei of bonded atoms
(c) $(2p_\pi - 3d_\pi)$ pi-bond is stronger than $(2p_\pi - 3p_\pi)$ pi-bond.
(d) σ -bond has primary effect of decide direction of covalent bond, while π -bond has no primary effect in direction of bond
- 42. (M)** In which of the following there is intermolecular hydrogen bonding ?
- (a) Water (b) Ethanol
(c) Acetic acid (d) H-F
- 43. (M)** Which of the following statements are correct about sulphur hexafluoride ?
- (a) all S - F bonds are equivalent
(b) SF_6 is a planar molecule
(c) oxidation number of sulphur is the same as number of electrons of sulphur involved in bonding
(d) sulphur has acquired the electronic structure of the gas argon
- 44. (M)** Ionic compounds in general do not possess :
- (a) high melting points and non-directional bonds
(b) high melting points and low-boiling points
(c) directional bonds and low-boiling points
(d) high solubilities in polar and non-polar solvents
- 45. (M)** Which of the following statements is incorrect ?
- (a) A σ bond is weaker than a π bond
(b) There are four co-ordinate bonds in the NH_4^+ ions
(c) The covalent bond is directional in nature
(d) HF is less polar than HCl
- 46. (M)** Correct order of decreasing boiling points is :
- (a) $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$
(b) $\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$
(c) $\text{Br}_2 > \text{Cl}_2 > \text{F}_2$
(d) $\text{CH}_4 > \text{GeH}_4 > \text{SiH}_4$
- 47. (M)** Which of the following statements is correct ?
- (a) ClF_3 molecule is bent 'T' shape
(b) In SF_4 molecule, F - S - F equatorial bond angle is 103° due to lp-lp repulsion
(c) In $[\text{ICl}_4]^-$ molecular ion, Cl - I - Cl bond angle is 90°
(d) In OBr_2 , the bond angle is less than OCl_2
- 48. (M)** Which of the following is (are) V-shaped ?
- (a) S_3^{2-} (b) I_3^-
(c) N_3^- (d) I_3^+
- 49. (M)** Which of the following are true ?
- (a) van der Waals forces are responsible for the formation of molecular crystals
(b) Branching lower the boiling points of isomeric organic compounds due to decrease in van der Waals forces
(c) In graphite, van der Waals forces act between the carbon layers
(d) In diamond, van der Waals forces act between the carbon layers
- 50. (M)** Which of the following statement is incorrect :
- (a) O_2 is paramagnetic, O_3 is also paramagnetic
(b) O_2 is paramagnetic, O_3 is diamagnetic
(c) B_2 is paramagnetic, C_2 is also paramagnetic
(d) Different observation is found in their bond length when $\text{NO} \rightarrow \text{NO}^+$ and $\text{CO} \rightarrow \text{CO}^+$

51. (M) Which of the following statements are not correct ?

- (a) All C – O bonds in CO_3^{2-} are equal but not in H_2CO_3
- (b) All C – O bonds in HCO_2^- are equal but not in HCO_2H
- (c) C – O bond length in HCO_2^- is longer than C – O bond length in CO_3^{2-}
- (d) C – O bond length in HCO_2^- and C – O bond length in CO_3^{2-} are equal

52. (M) Which of the following is true for N_2O ?

- (a) Its molecule is linear
- (b) Symmetric N – O – N is a favoured structure as compared to N – N – O skeleton
- (c) Bond orders are fractional for N – N and N – O bonds
- (d) It is a neutral oxide

53. (M) If N_B is the number of bonding electrons and N_A is the number of antibonding electrons of a molecule. Then choose the incorrect statement(s) for the relationship, $N_B > N_A$:

- (a) Molecule may be stable or unstable
- (b) Molecule may have any integral, fractional or zero value of bond order
- (c) Molecule is only paramagnetic species
- (d) Molecule does not exist

54. (M) Correct order of B.pt. is/are :

- (a) $\text{H}_2 < \text{He}$ (b) $\text{H}_2 < \text{D}_2$
- (c) $\text{H}_2\text{O} < \text{D}_2\text{O}$ (d) $\text{NH}_3 < \text{SbH}_3$

55. (A) **Assertion :** In SO_3 molecule bond dissociation energy of all S = O bonds are not equivalent.

Reason : SO_3 molecule is having two types of $2p\pi - 3p\pi$ and $2p\pi - 3d\pi$ pi-bonds.

- (a) A (b) B
- (c) C (d) D

56. (A) **Assertion :** Water is a good solvent for ionic compounds but poor for covalent compounds.

Reason : Hydration energy of ions releases sufficient energy to overcome lattice energy and break hydrogen bonds in water while covalent compounds interact so weakly that even van der Waal's forces between molecules of covalent compounds cannot be broken.

- (a) A (b) B
- (c) C (d) D

57. (A) **Assertion :** $\text{H}\hat{\text{O}}\text{F}$ bond angle is higher than $\text{H}\hat{\text{O}}\text{Cl}$.

Reason : Oxygen is more electronegative than chlorine.

- (a) A (b) B
- (c) C (d) D

58. (A) **Assertion :** $(\text{CH}_3)_3\text{N}$ geometry is pyramidal but in case $(\text{SiH}_3)_3\text{N}$ it is planar.

Reason : The maximum covalency of Si is six but that of C is four.

- (a) A (b) B
- (c) C (d) D

59. (A) **Assertion :** The HF_2^- ion exists in the solid state & also in liquid state but not in aqueous state.

Reason : The magnitude of hydrogen bonds among HF molecules is weaker than that in between HF and H_2O .

- (a) A (b) B
- (c) C (d) D

Comprehension

Ionic bond is defined as the electrostatics force of attraction holding the oppositely charged ions. Ionic compounds are mostly crystalline solid having high melting and boiling points, electrical conductivity in molten state, solubility in water etc. Covalent bond is defined as the force which binds atoms of same or different elements by mutual sharing of electrons in a covalent bond. Covalent compounds are solids, liquids or gases. They have low melting and boiling points compounds. They are more soluble in non-polar solvents.

60. (C) The valence electrons are involved in formation of covalent bonds is/are called :

- (a) non-bonding electrons
- (b) lone pairs
- (c) unshared pairs
- (d) none of these

61. (C) The amount of energy released when one mole of ionic solid is formed by packing of gaseous ion is called :

- (a) Ionisation energy (b) Solvation energy
- (c) Lattice energy (d) Hydration energy

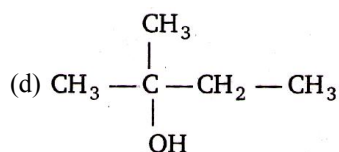
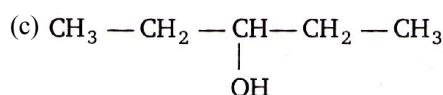
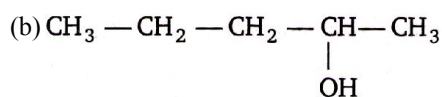
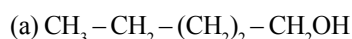
62. (C) Which of the following is arranged order of increasing boiling point ?

- (a) $\text{H}_2\text{O} < \text{CCl}_4 < \text{CS}_2 < \text{CO}_2$
- (b) $\text{CO}_2 < \text{CS}_2 < \text{CCl}_4 < \text{H}_2\text{O}$
- (c) $\text{CS}_2 < \text{H}_2\text{O} < \text{CO}_2 < \text{CCl}_4$
- (d) $\text{CCl}_4 < \text{H}_2\text{O} < \text{CO}_2 < \text{CS}_2$

Comprehension

In general boiling point of covalent compounds is effected due to increasing molecular weight and hydrogen bonding. Thus it is observed that the boiling point of octanol is greater than water while that of methanol is lesser than water. The other properties which are considerably affected due to hydrogen bond formation are acidic nature, melting point, anomalous behaviour of water below 4°C. It is therefore concluded that hydrogen bonding plays a vital role in explaining many observable facts.

63. (C) Which among the following has maximum boiling point ?



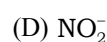
64. (C) Which of the following statements is true ?

- (a) the lattice structure of ice involves true covalent bond
- (b) the lattice structure of ice is the result of dipole-dipole interaction
- (c) the lattice structure of ice is the result of intra and inter-molecular hydrogen bond formation
- (d) the lattice structure of ice is only due to inter-molecular hydrogen bonding

65. (C) K_{a2} of Maleic acid is lesser than K_{a2} of fumaric acid due to:

- (a) intramolecular hydrogen bonding in the ion formed after one proton removal in fumaric acid
- (b) intermolecular hydrogen bonding in the ion formed after one proton removal in maleic acid
- (c) intramolecular hydrogen bonding in the ion formed after one proton removal in maleic acid
- (d) intermolecular hydrogen bonding in the ion formed after one proton removal in fumaric acid

66. (X) Column I



Column II

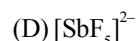
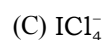
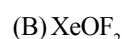
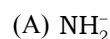
(P) Largest bond angle

(Q) Lowest bond angle

(R) sp^2 -hybridisation

(S) sp^3 -hybridisation

67. (X) Column I



Column II

(P) Square pyramidal

(Q) V-shaped

(R) T-shaped

(S) Square planar

68. (X) Column – I

(Axial/sideways

combination of appropriate/
inappropriate pure orbitals)

Column – II

(Types of molecular
orbitals)

(A) $p + p$ pure orbitals

(B) $s + p$ pure orbitals

(C) (non-axial) $d + p$
pure orbitals

(D) (axial) $d + p$ pure
orbitals

(P) σ -bonding molecular

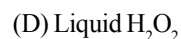
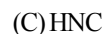
(Q) π -anti-bonding molecular
orbital

(R) σ -anti-bonding molecular
orbital

(S) π -bonding molecular orbital

(T) Non-bonding molecular
orbital

69. (X) Column – I



Column – II

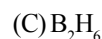
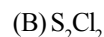
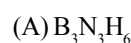
(P) Hydrogen bond

(Q) Co-ordinate bond

(R) Ionic bond

(S) Covalent bond

70. (X) Column – I



Column – II

(P) Planar geometry

(Q) Non-polar geometry

(R) No lone pair

(S) Non-polar molecule

- 71. (X)** **Column – I** **Column – II**
- | (Species) | (Bond angle) |
|---------------------|-----------------|
| (A) NO_2^+ | (P) 180° |
| (B) NO_2^- | (Q) 120° |
| (C) NO_2 | (R) 134° |
| (D) NO_3^- | (S) 115° |
| | (T) 109° |
- 72. (I)** Among the following, total no. of planar species is :
- | | |
|------------------------|------------------------|
| (i) SF_4 | (ii) BrF_3 |
| (iii) XeF_2 | (iv) IF_5 |
| (v) SbF_4^- | (vi) SF_5^- |
| (vii) SeF_3^+ | (viii) CH_3^+ |
| (ix) PCl_4^+ | |
- 73. (I)** Consider the following species :
- | | |
|--|--|
| (i) CH_3^+ | (ii) $(\text{C}_3\text{H}_5)_3\text{Al}$ |
| (iii) HCHO | (iv) CH_4 |
| (v) $(\text{C}_2\text{H}_5)_3\text{N}$ | (vi) TiCl_4 |
| (vii) CO_2 | (viii) SiCl_4 |
| (ix) BF_3 | |
- Then find out total number of species which can act as Lewis acid.

EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTION

Objective Question - I (Only one correct option)

- The compound which contains both ionic and covalent bonds is (1979)
 - CH_4
 - H_2
 - KCN
 - KCl
- The total number of electrons that taken part in forming the bonds in N_2 is (1980)
 - 2
 - 4
 - 6
 - 10
- Which of the following compound is covalent ? (1980)
 - H_2
 - CaO
 - KCl
 - Na_2S
- Element X is strongly electropositive and element Y is strongly electronegative. Both are univalent. The compound formed would be (1980)
 - X^+Y^-
 - X^-Y^+
 - $\text{X}-\text{Y}$
 - $\text{X} \rightarrow \text{Y}$
- Which of the following is soluble in water (1980)
 - CS_2
 - $\text{C}_2\text{H}_5\text{OH}$
 - CCl_4
 - CHCl_3
- If a molecule MX_3 has zero dipole moment, the sigma bonding orbitals used by M (atomic number < 21) are (1981)
 - Pure p
 - sp hybridised
 - sp^2 hybridised
 - sp^3 hybridised
- Among the following, the molecule that is linear is (1982)
 - CO_2
 - NO_2
 - SO_2
 - ClO_2
- The ion that is isoelectronic with CO is (1982)
 - CN^-
 - O_2^+
 - O_2^-
 - N_2^+
- Carbon tetrachloride has no net dipole moment because of (1983)
 - its planar structure
 - its regular tetrahedral structure
 - similar sizes of carbon and chlorine
 - similar electron affinities of carbon and chlorine
- Which one among the following does not have the hydrogen bond ? (1983)
 - Phenol
 - Liquid NH_3
 - Water
 - HCl
- On hybridisation of one s and one p-orbital we get (1984)
 - two mutually perpendicular orbitals
 - two orbitals at 180°
 - four orbitals directed tetrahedrally
 - three orbitals in a plane
- The bond between two identical non-metal atoms has a pair of electrons (1986)
 - unequally shared between the two
 - transferred fully from one atom to another
 - with identical spins
 - equally shared between them
- The hybridisation of sulphur in sulphur dioxide is (1986)
 - sp
 - sp^3
 - sp^2
 - dsp^2
- Of the following compounds, which will have a zero dipole moment ? (1987)
 - 1, 1-dichloroethylene
 - cis-1, 2-dichloroethylene
 - trans-1, 2-dichloroethylene
 - None of the above
- The molecule that has linear structure is (1988)
 - CO_2
 - NO_2
 - SO_2
 - SiO_2
- The $\text{Cl}-\text{C}-\text{Cl}$ angle in 1, 1, 2, 2-tetrachloroethene and tetrachloromethane respectively will be about (1988)
 - 120° and 109.5°
 - 90° and 109.5°
 - 109° and 90°
 - 109.5° and 120°
- The species in which the central atom uses sp^2 hybrid orbitals in its bonding is (1988)
 - PH_3
 - NH_3
 - CH_3^+
 - SbH_3
- The compound in which C^* uses its sp^3 hybrid orbitals for bond formation is : (1989)
 - HCOOH
 - $(\text{H}_2\text{N})_2\text{C}^*\text{O}$
 - $(\text{CH}_3)_3\text{C}^*\text{OH}$
 - $\text{CH}_3\text{C}^*\text{HO}$
- The molecule which has zero dipole moment is (1989)
 - CH_2Cl_2
 - BF_3
 - NF_3
 - ClO_2

20. Which of the following is paramagnetic ? (1989)
(a) O_2^- (b) CN^-
(c) CO (d) NO^+
21. The molecule which has pyramidal shape is (1989)
(a) PCl_3 (b) SO_3
(c) CO_3^{2-} (d) NO_3^-
22. The number of hybrid orbitals used by the chlorine atom in ClO_2^- is (1992)
(a) 2 (b) 4
(c) 3 (d) 1
23. The type of hybrid orbitals used by the chlorine atom in ClO_2^- is (1992)
(a) sp^3 (b) sp^2
(c) sp (d) None of these
24. The maximum possible number of hydrogen bonds a water molecule can form is (1992)
(a) 2 (b) 4
(c) 3 (d) 1
25. Pick out the isoelectronic structures from the following: (1993)
I. CH_3^+ II. H_3O^+
III. NH_3 IV. CH_3^-
(a) I and II (b) III and IV
(c) I and III (d) II, III and IV
26. Number of paired electrons in O_2 molecule is : (1995)
(a) 7 (b) 8
(c) 16 (d) 14
27. Which one of the following molecules is planar ? (1996)
(a) NF_3 (b) NCl_3
(c) PH_3 (d) BF_3
28. Among the following species, identify the isostructural pairs. (1996)
 NF_3 , NO_3^- , BF_3 , H_3O^+ , HN_3
(a) $[NF_3, NO_3^-]$ and $[BF_3, H_3O^+]$
(b) $[NF_3, NO_3^-]$ and $[NO_3^-, BF_3]$
(c) $[NF_3, H_3O^+]$ and $[NO_3^-, BF_3]$
(d) $[NF_3, H_3O^+]$ and $[HN_3, BF_3]$
29. The number and type of bonds between two carbon atoms in CaC_2 are (1996)
(a) one sigma (σ) and one pi (π) bonds
(b) one sigma (σ) and two pi (π) bonds
(c) one sigma (σ) and one half pi (π) bonds
(d) one sigma (σ) bond
30. Arrange the following compounds in order of increasing dipole moment, toluene (I), m-dichlorobenzene (II), o-dichlorobenzene (III), p-dichlorobenzene (IV) (1996)
(a) $I < IV < II < III$ (b) $IV < I < II < III$
(c) $IV < I < III < II$ (d) $IV < II < I < III$
31. Which contains both polar and non-polar bonds ? (1997)
(a) NH_4Cl (b) HCN
(c) H_2O_2 (d) CH_4
32. The critical temperature of water is higher than that of O_2 because the H_2O molecule has (1997)
(a) fewer electrons than O_2
(b) two covalent bonds
(c) V-shape (d) dipole moment.
33. Among KO_2 , AlO_2^- , BaO_2 and NO_2^+ , unpaired electron is present in (1997)
(a) NO_2^+ and BaO_2 (b) KO_2 and AlO_2^-
(c) KO_2 only (d) BaO_2 only
34. Which one of the following compounds has sp^2 -hybridisation ?
(a) CO_2 (b) SO_2
(c) N_2O (d) CO
35. The geometry and the type of hybrid orbital present about the central atom in BF_3 is (1998)
(a) linear, sp (b) trigonal planar, sp^2
(c) tetrahedral, sp^3 (d) pyramidal, sp^3
36. The geometry of H_2S and its dipole moment are (1999)
(a) angular and non-zero (b) angular and zero
(c) linear and non-zero (d) linear and zero
37. In compounds of type ECl_3 , where E = B, P, As or Bi, the angles $Cl-E-Cl$ (1999)
(a) $B > P = As = Bi$ (b) $B > P > As > Bi$
(c) $B < P = As = Bi$ (d) $B < P < As < Bi$
38. In the compound $CH_2 = CH-CH_2-CH_2-C \equiv CH$, the C_2-C_3 bonds is of
(a) $sp-sp^2$ (b) sp^3-sp^3
(c) $sp-sp^3$ (d) sp^2-sp^3

39. The correct order of increasing C – O bond length of CO , CO_3^{2-} , CO_2 , is (1999)
- (a) $\text{CO}_3^{2-} < \text{CO}_2 < \text{CO}$ (b) $\text{CO}_2 < \text{CO}_3^{2-} < \text{CO}$
- (c) $\text{CO} < \text{CO}_3^{2-} < \text{CO}_2$
- (d) $\text{CO} < \text{CO}_2 < \text{CO}_3^{2-}$
40. Molecular shapes of SF_4 , CF_4 and XeF_4 are (2000)
- (a) the same, with 2, 0 and 1 lone pairs of electrons respectively
- (b) the same, with 1, 1, and 1 lone pairs of electrons respectively
- (c) different, with 0, 1 and 2 lone pairs of electrons respectively
- (d) different, with 1, 0 and 2 lone pairs of electrons respectively.
41. The hybridisation of atomic orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ are (2000)
- (a) sp , sp^3 and sp^2 respectively
- (b) sp , sp^2 and sp^3 respectively
- (c) sp^2 , sp and sp^3 respectively
- (d) sp^2 , sp^3 and sp respectively
42. Amongst H_2O , H_2S , H_2Se and H_2Te , the one with the highest boiling point is (2000)
- (a) H_2O because of hydrogen bonding
- (b) H_2Te because of higher molecular weight
- (c) H_2S because of hydrogen bonding
- (d) H_2Se because of lower molecular weight
43. The common features among the species CN^- , CO and NO^+ are (2001)
- (a) bond order three and isoelectronic
- (b) bond order three and weak field ligands
- (c) bond order two and acceptors
- (d) isoelectronic and weak field ligands
44. The correct order of hybridisation of the central atom in the following species NH_3 , $[\text{PtCl}_4]^{2-}$, PCl_5 and BCl_3 is
- (a) dsp^2 , dsp^3 , sp^2 and sp^3
- (b) sp^3 , dsp^2 , sp^3d , and sp^2
- (c) dsp^2 , sp^2 , sp^3 , dsp^3
- (d) dsp^2 , sp^3 , sp^2 , dsp^3
45. Specify the coordination geometry around and hybridisation of N and B atoms in a 1 : 1 complex of BF_3 and NH_3 (2002)
- (a) N : tetrahedral, sp^3 ; B : tetrahedral, sp^3
- (b) N : pyramidal, sp^3 ; B : pyramidal, sp^3
- (c) N : pyramidal, sp^3 ; B : planar, sp^2
- (d) N : pyramidal, sp^3 ; B : tetrahedral, sp^3
46. The nodal plane in the π -bond of ethene is located in (2002)
- (a) the molecular plane
- (b) a plane parallel to the molecular plane
- (c) a plane perpendicular to the molecular plane which bisects the carbon σ -bond at right angle
- (d) a plane perpendicular to the molecular plane which contains the carbon-carbon σ -bond
47. Which of the following molecular species has unpaired electrons (s) ? (2002)
- (a) N_2 (b) F_2
- (c) O_2^- (d) O_2^{2-}
48. Among the following the molecular with the highest dipole moment is
- (a) CH_3Cl (b) CH_2Cl_2
- (c) CHCl_3 (d) CCl_4
49. Which of the following are isoelectronic and isostructural ? (2003)
- NO_3^- , CO_3^{2-} , ClO_3^- , SO_3
- (a) NO_3^- , CO_3^{2-} (b) SO_3 , NO_3^-
- (c) ClO_3^- , CO_3^{2-} (d) CO_3^{2-} , SO_3
50. Number of lone pair (s) in XeOF_4 is/are (2004)
- (a) 0 (b) 1
- (c) 2 (d) 3
51. According to MO theory, (2004)
- (a) O_2^+ is paramagnetic and bond order greater than O_2
- (b) O_2^+ is paramagnetic and bond order less than O_2
- (c) O_2^+ is diamagnetic and bond order is less than O_2
- (d) O_2^+ is diamagnetic and bond order is more than O_2
52. Which species has the maximum number of lone pair of electrons on the central atom ? (2005)
- (a) $[\text{ClO}_3]^-$ (b) XeF_4
- (c) SF_4 (d) $[\text{I}_3]^-$

53. Among the following, the paramagnetic compound is (2007)
- (a) Na_2O_2 (b) O_3
(c) N_2O (d) KO_2
54. The species having bond order different from that in CO is (2007)
- (a) NO^- (b) NO^+
(c) CN^- (d) N_2
55. Assuming that Hund's rule is violated, the bond order and magnetic nature of the diatomic molecule B_2 is (2010)
- (a) 1 and diamagnetic (b) 0 and diamagnetic
(c) 1 and paramagnetic (d) 0 and paramagnetic
56. The species having pyramidal shape is (2010)
- (a) SO_3 (b) BrF_3
(c) SiO_3^{2-} (d) OSF_2
57. In allene (C_3H_4), the type (s) of hybridization of the carbon atoms, is (are) (2012)
- (a) sp and sp^3 (b) sp and sp^2
(c) only sp^3 (d) sp^2 and sp^3
58. The shape of XeO_2F_2 molecule is
- (a) trigonal bipyramidal (b) square planar
(c) tetrahedral (d) see-saw
59. Assuming 2s-2p mixing is NOT operative, the paramagnetic species among the following is (2014)
- (a) Be_2 (b) B_2
(c) C_2 (d) N_2

Objective Question - II

(One or more than one correct option)

60. Dipole moment is shown by (1986)
- (a) 1, 4-dichlorobenzene (b) cis-1, 2-dichloroethene
(c) trans-1, 2-dichloroethene
(d) trans-1, 2-dichloro-2-pentene
61. CO_2 is isostructural with : (1986)
- (a) HgCl_2 (b) SnCl_2
(c) C_2H_2 (d) NO_2
62. The linear structure is assumed by : (1991)
- (a) SnCl_2 (b) NCO^-
(c) CS_2 (d) NO_2^+
63. Which of the following have identical bond order ? (1992)
- (a) CN^- (b) O_2^-
(c) NO^+ (d) CN^+
64. The molecules that will have dipole moment are (1992)
- (a) 2, 2-dimethyl propane (b) trans-2-pentene
(c) cis-3-hexene
(d) 2, 2, 3, 3-tetramethyl butane
65. The hyperconjugative stabilities of tert-butyl cation and 2-butene, respectively, are due to (2005)
- (a) $\sigma \rightarrow \text{p}$ (empty) and $\sigma \rightarrow \pi^*$ electron delocalisations
(b) $\sigma \rightarrow \sigma^*$ and $\sigma \rightarrow \pi$ electron delocalisations
(c) $\sigma \rightarrow \text{p}$ (filled) and $\sigma \rightarrow \pi$ electron delocalisations
(d) p (filled) $\rightarrow \sigma^*$ and $\sigma \rightarrow \pi^*$ electrons delocalisations
66. According to Molecular Orbital Theory. (2016)
- (a) C_2^{2-} is expected to be diamagnetic
(b) O_2^{2+} is expected to have longer bond length than O_2
(c) N_2^+ and N_2^- have the same bond order
(d) He_2^+ has the same energy as two isolated He atoms.
67. The compound(s) with TWO lone pairs of electrons on the central atom is (are)
- (a) BrF_5 (b) ClF_3
(c) XeF_4 (d) SF_4

ASSERTION & REASON

68. **Assertion :** LiCl is predominantly a covalent compound.
Reason : Electronegativity difference between Li and Cl is too small.
- (A) If both ASSERTION and REASON are true and reason is the correct explanation of the assertion.
(B) If both ASSERTION and REASON are true but reason is not the correct explanation of the assertion.
(C) If ASSERTION is true but REASON is false.
(D) If ASSERTION is false but REASON is true.
(E) If both ASSERTION and REASON are false.

Match the Columns

69. Match the reactions in Column-I with nature of the reactions/type of the products in Column-II. (2007)
- | Column-I | Column-II |
|--|---|
| (A) $\text{O}_2^- \rightarrow \text{O}_2 + \text{O}_2^{2-}$ | (p) Redox reaction |
| (B) $\text{CrO}_4^{2-} + \text{H}^+ \rightarrow$ | (q) One of the products has trigonal planar structure |
| (C) $\text{MnO}_4^- + \text{NO}_2^- + \text{H}^+ \rightarrow$ | (r) Dimeric bridged tetrahedral metal ion |
| (D) $\text{NO}_3^- + \text{H}_2\text{SO}_4 + \text{Fe}^{2+} \rightarrow$ | (s) Disproportionation |

70. Match each of the diatomic molecules in Column-I with its property/properties in Column-II. (2009)

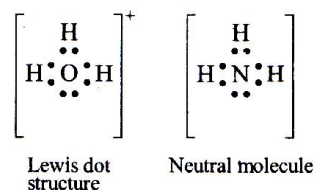
Column - I	Column-II
(A) B_2	(p) Paramagnetic
(B) N_2	(q) Undergoes oxidation
(C) O_2^-	(r) Undergoes reduction
(D) O_2	(s) Bond order ≥ 2
	(t) Mixing of 's' and 'p' orbitals

Fill in the Blanks

71. The angle between two covalent bonds is maximum in (1981)
72. Pair of molecules which forms strongest intermolecular hydrogen bonds is
(SiH_4 and SiF_4 , acetone and $CHCl_3$, formic acid and acetic acid) (1981)
73. There are π bonds in a nitrogen molecule. (1982)
74. hybrid orbitals of nitrogen atom are involved in the formation of ammonium ion. (1982)
75. The shape of $[CH_3]^+$ is (1990)
76. The valence atomic orbitals on C in silver acetylide is hybridized. (1990)
77. The kind of delocalization involving sigma bond orbitals is called (1994)
78. The two types of bonds present in B_2H_6 are covalent and (1994)
79. When N_2 goes to N_2^+ , the N—N bond distance, and when O_2 goes to O_2^+ the O—O bond distance (1996)
80. Among N_2O , SO_2 , I_3^+ and I_3^- , the linear species are and (1997)
85. sp^3 hybrid orbitals have equal s and p character. (1987)
86. The presence of polar bonds in a polyatomic molecule suggests that the molecule has non-zero dipole-moment. (1990)
87. H_2O molecule is linear. (1993)
88. The dipole moment of CH_3F is greater than that of CH_3Cl . (1993)

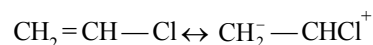
Subjective Questions

89. Arrange the following :
- (i) N_2 , O_2 , F_2 , Cl_2 in increasing order of bond dissociation energy. (1988)
- (ii) Increasing strength of hydrogen bonding in halogen acids (HX). (1991)
- (iii) In the decreasing order of the O—O bond length present in them (2004)
- O_2 , KO_2 and $O_2 [AsF_4]$
90. Using VSEPR theory, draw the shape of PCl_5 and BrF_5 . (2003)
91. State four major physical properties that can be used to distinguish between covalent and ionic compounds. Mention the distinguish features in each case. (1978)
92. Write the Lewis dot structural formula for each of the following. Give also, the formula of a neutral molecule, which has the same geometry and the same arrangement of the bonding electrons as in each of the following. An example is given below in the case H_3O^+ and NH_3 . (1983)



True/False

81. Linear overlap of two atomic p-orbitals leads to a sigma bond. (1983)
82. All molecules with polar bonds have dipole moment. (1985)
83. $SnCl_2$ is a non-linear molecule. (1985)
84. In benzene, carbon uses all the three p-orbitals for hybridisation. (1987)
- (i) O_2^{2-} (ii) CO_3^{2-}
- (iii) CN^- (iv) NCS^-
93. What effect should the following resonance of vinyl chloride have on its dipole moment ? (1987)



94. Arrange the following as stated.
“Increasing strength of hydrogen bonding (X–H–X).”
O, S, F, Cl, N (1991)
95. Give reasons in two or three sentences only for the following :
“Hydrogen peroxide acts as an oxidizing as well as a reducing agent.” (1992)
96. The dipole moment of KCl is 3.336×10^{-29} C-m which indicates that it is highly polar molecule. The inter atomic distance between K^+ and Cl^- in this molecule is 2.6×10^{-10} m. Calculate the dipole moment of KCl molecule if there were opposite charges of one fundamental unit located at each nucleus. Calculate the percentage ionic character of KCl. (1993)
97. Explain the difference in the nature of bonding in LiF and LiI. (1996)
98. Using the VSEPR theory, identify the type of hybridisation and draw the structure of OF_2 . What are the oxidation states of O and F ? (1997)
99. In the reaction, $I^- + I_2 \longrightarrow I_3^-$, which is the Lewis acid ? (1997)
100. Between Na^+ and Ag^+ , which is stronger Lewis acid and why ? (1997)
101. (a) Arrange the following ions in order of their increasing radii : Li^+ , Mg^{2+} , K^+ , Al^{3+} (1997)
(b) Arrange the following sulphates of alkaline earth metals in order of decreasing thermal stability : $BeSO_4$, $MgSO_4$, $CaSO_4$, $SrSO_4$.
102. Interpret the non-linear shape of H_2S molecule and non-planar shape of PCl_3 using valence shell electron pair repulsion (VSEPR) theory. (Atomic number : H = 1, P = 15, S = 16, Cl = 17) (1998)
103. (a) Write the MO electron distribution of O_2 . Specify its bond order and magnetic property. (2000)
(b) Draw the molecular structures of XeF_2 , XeF_4 and XeO_2F_2 , indicating the location of lone pair(s) of electrons. (2000)
104. Draw the shape of XeF_4 and OSF_4 according to VSEPR theory. Show the lone pair of electrons on the central atom. (2004)
105. On the basis of ground state electronic configuration arrange the following molecules in increasing O–O bond length order. KO_2 , O_2 , $O_2[AsF_6]$ (2004)
106. Predict whether the following molecules are iso-structural or not. Justify your answer.
(i) NMe_3 (ii) $N(SiMe_3)_3$ (2005)
107. The total number of lone pairs of electrons in N_2O_3 is (2015)
108. The sum of the number of lone pairs of electrons on each central atom in the following species is
 $[TeBr_6]^{2-}$, $[BrF_2]^+$, SNF_3 and $[XeF_3]^-$
(Atomic numbers : N = 7, F = 9, S = 16, Br = 35, Te = 52, Xe = 54) (2017)
109. Among H_2 , He_2^+ , Li_2 , Be_2 , B_2 , C_2 , N_2 , O_2^- and F_2 , the number of diamagnetic species is
(Atomic numbers, H = 1, He = 2, Li = 3, Be = 4, B = 5, C = 6, N = 7, O = 8, F = 9) (2017)

ANSWER KEY

EXERCISE - 1 : (Basic Objective Questions)

1. (d)	2. (a)	3. (a)	4. (d)	5. (b)	6. (a)	7. (b)	8. (c)
9. (b)	10. (d)	11. (c)	12. (d)	13. (c)	14. (c)	15. (a)	16. (d)
17. (a)	18. (a)	19. (d)	20. (a)	21. (a)	22. (a)	23. (c)	24. (c)
25. (a)	26. (d)	27. (a)	28. (c)	29. (c)	30. (c)	31. (d)	32. (a)
33. (c)	34. (d)	35. (a)	36. (c)	37. (b)	38. (a)	39. (b)	40. (b)
41. (d)	42. (d)	43. (c)	44. (a)	45. (b)	46. (d)	47. (a)	48. (d)
49. (d)	50. (c)	51. (a)	52. (b)	53. (b)	54. (a)	55. (b)	56. (b)
57. (d)	58. (d)	59. (c)	60. (c)	61. (a)	62. (a)	63. (b)	64. (d)
65. (a)	66. (b)	67. (c)	68. (b)	69. (d)	70. (d)	71. (d)	72. (b)
73. (c)	74. (d)	75. (b)	76. (a)	77. (b)	78. (d)	79. (a)	80. (a)
81. (d)	82. (a)	83. (a)	84. (a)	85. (b)	86. (a)	87. (d)	88. (a)
89. (c)	90. (b)	91. (b)	92. (a)	93. (b)	94. (d)		

EXERCISE - 2 : (PREVIOUS YEAR JEE MAINS QUESTIONS)

1. (a)	2. (b)	3. (d)	4. (b)	5. (d)	6. (a)	7. (c)	8. (a)
9. (b)	10. (c)	11. (c)	12. (b)	13. (b)	14. (b)	15. (c)	16. (c)
17. (d)	18. (a)	19. (a)	20. (c)	21. (d)	22. (c)	23. (c)	24. (a)
25. (c)	26. (a)	27. (b)	28. (b)	29. (d)	30. (b)	31. (a)	32. (d)
33. (d)	34. (b)	35. (c)	36. (c)	37. (c)	38. (b)	39. (c)	40. (a)
41. (c)	42. (d)	43. (b)	44. (d)	45. (d)	46. (a)		

JEE Mains Online

1. (b)	2. (d)	3. (d)	4. (c)	5. (a)	6. (b)	7. (c)	8. (c)
9. (b)	10. (a)	11. (d)	12. (d)	13. (b)	14. (b)	15. (c)	16. (b)
17. (d)	18. (c)	19. (c)	20. (d)				

EXERCISE - 3 : (Advanced Objective Questions)

1. (c)	2. (c)	3. (b)	4. (b)	5. (b)	6. (b)	7. (c)	8. (a)
9. (b)	10. (a)	11. (b)	12. (b)	13. (a)	14. (b)	15. (b)	16. (d)
17. (d)	18. (c)	19. (b)	20. (c)	21. (a)	22. (a)	23. (c)	24. (c)
25. (c)	26. (d)	27. (b)	28. (b)	29. (c)	30. (c)	31. (a)	32. (c)
33. (b)	34. (d)	35. (b)	36. (a)	37. (b)	38. (b)	39. (c)	40. (a)
41. (a,b,c,d)	42. (a,b,c,d)	43. (a,c)	44. (b,c,d)	45. (a,b,d)	46. (a,b,c)	47. (a,c)	48. (a,d)
49. (a,b,c)	50. (a,c,d)	51. (c,d)	52. (a,c,d)	53. (b,c,d)	54. (b,c)	55. (a)	56. (c)

57. (b) 58. (a) 59. (c) 60. (d) 61. (c) 62. (b) 63. (a) 64. (d)
65. (c) 66. $A \rightarrow R$; $B \rightarrow P$; $C \rightarrow Q, S$; $D \rightarrow R$ 67. $A \rightarrow Q$; $B \rightarrow R$; $C \rightarrow S$; $D \rightarrow P$
68. $A \rightarrow P, Q, R, S, T$; $B \rightarrow P, R, T$; $C \rightarrow Q, S, T$; $D \rightarrow P, R, T$ 69. $A \rightarrow Q, R, S$; $B \rightarrow P, Q, R, S$; $C \rightarrow P, S$; $D \rightarrow P, S$
70. $A \rightarrow P, R, S$; $B \rightarrow Q$; $C \rightarrow Q, R, S$; $D \rightarrow P, S$ 71. $A \rightarrow P$; $B \rightarrow S$; $C \rightarrow R$; $D \rightarrow Q$ 72. (0003) 73. (0007)

EXERCISE - 4 : (Previous Year JEE Advanced Questions)

1. (c) 2. (c) 3. (a) 4. (a) 5. (b) 6. (c) 7. (a) 8. (a)
9. (b) 10. (d) 11. (b) 12. (d) 13. (c) 14. (c) 15. (a) 16. (a)
17. (c) 18. (c) 19. (b) 20. (a) 21. (a) 22. (b) 23. (a) 24. (b)
25. (d) 26. (d) 27. (d) 28. (c) 29. (b) 30. (b) 31. (c) 32. (d)
33. (c) 34. (b) 35. (b) 36. (a) 37. (b) 38. (d) 39. (d) 40. (d)
41. (b) 42. (a) 43. (a) 44. (b) 45. (a) 46. (a) 47. (c) 48. (a)
49. (a) 50. (b) 51. (a) 52. (d) 53. (d) 54. (a) 55. (a) 56. (d)
57. (b) 58. (d) 59. (c) 60. (b,d) 61. (a,c) 62. (b, c, d) 63. (a, c) 64. (b,c,d)
65. (b, c) 66. (a,c) 67. (b,c) 68. (c) 69. A-p,s; B-r; C-p,q; D-p.
70. A-p,q,r,t; B-q, r, s, t; C-p, q, r, t; D-p, r, s, t. 71. CO_2 72. HCOOH and CH_3COOH 73. 2
74. sp^3 75. planar 76. sp 77. hyperconjugation 78. three centre bond two electron.
79. increases, decreases 80. N_2O , I_3^- 81. (T) 82. (F) 83. (T) 84. (F)
85. (F) 86. (F) 87. (F) 88. (T)
89. (i) $\text{F}_2 < \text{Cl}_2 < \text{O}_2 < \text{N}_2$; (ii) $\text{S} < \text{Cl} < \text{N} < \text{O} < \text{F}$; (iii) $\text{KO}_2 > \text{O}_2 > \text{O}_2 [\text{AsF}_4]$

Dream on !!

